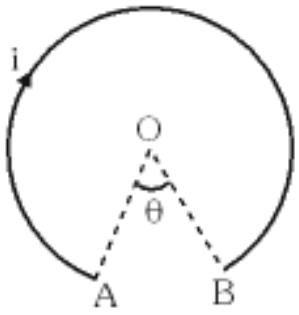
PHYSICS

1) A circular loop of radius r is carrying a current I . The ratio of magnetic field at the centre of circular loop and at a distance r from the center of the loop on its axis is :

(1) $1 : 3\sqrt{2}$
(2) $3\sqrt{2} : 2$
(3) $2\sqrt{2} : 1$
(4) $1 : \sqrt{2}$

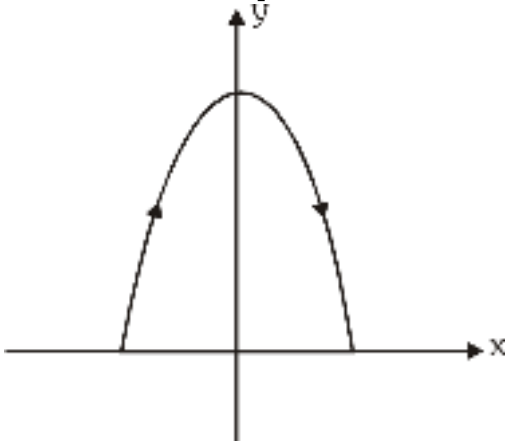
2) A current carrying wire AB of the length 2πR is turned along a circle, as shown in figure. The



magnetic field at the centre O.

(1) $\frac{\mu_0 i}{2R}\left(\frac{2\pi-\theta}{2\pi}\right)^2$
(2) $\frac{\mu_0 i}{2R}\left(\frac{2\pi-\theta}{2\pi}\right)$
(3) $\frac{\mu_0 i}{2R}(2\pi-\theta)$
(4) $\frac{\mu_0 i}{2R}(2\pi+\theta)^2$

3) A wire carrying a current of 4A is bent in the form of a parabola $x^2 + y = 16$ as shown in figure, where x and y are in meter. The wire is placed in a uniform magnetic field $\vec{B} = 5\hat{k}$ tesla. The force



acting on the wire is

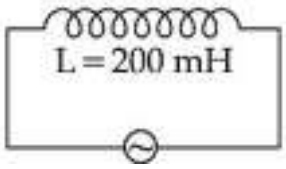
(1) $80\hat{j}$ N

(2) $-80\hat{j}$ N

(3) $-160\hat{j}$ N

(4) $160\hat{j}$ N

4) As shown in the figure an inductor of inductance 200 mH is connected to an AC source of emf 220 V and frequency 50 Hz. The instantaneous voltage of the source is 0 V when the peak value of current is $\frac{\sqrt{a}}{\pi}$A. The value of a is _____.



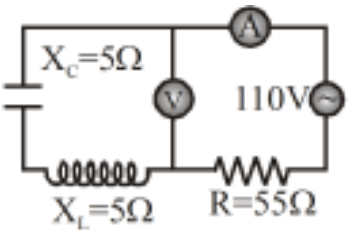
(1) 142

(2) 242

(3) 300

(4) 175

5) The reading of voltmeter (in volt) in the circuit shown will be :-



(1) 110 V

(2) 0 V

(3) 55 V

(4) 20 V

6) Expression for radiation pressure of EMW on perfectly absorbing surface is given by

(1) $\frac{1}{2}\in_0 E_0^2$

(2) $\frac{B_0^2}{\mu_0}$

(3) $\frac{I}{2C}$

(4) $\frac{2I}{C}$

7) Match List-I with List-II.

	List-I		**List-II**
A	Microwaves	I	Physiotherapy
B	UV rays	II	Treatment of cancer

C	Infra-red rays	III	Lasik eye surgery
D	X-rays	IV	Aircraft navigation

Choose the correct answer from the option given below:

(1) A-II, B-IV, C-III, D-I

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-III, C-I, D-II

(4) A-III, B-II, C-I, D-IV

8) A small coil of 10 turns is placed inside a solenoid of length 20 cm and 240 turns, carry a current of $\frac{10}{\pi}$A. The area of small coil is 2.5 cm^2 and resistance 4.8 Ω then the current reduces to zero in 25 ms, the value of average induced current is :-

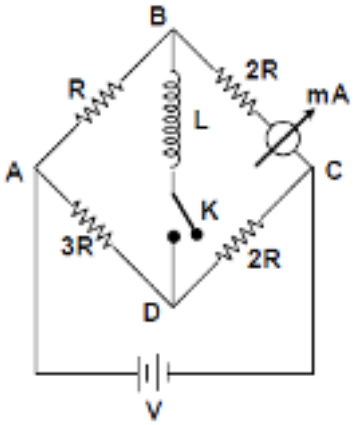
(1) 1 mA

(2) 2 mA

(3) 0.1 mA

(4) 0.4 mA

9) The reading of the ammeter, used in the electrical network shown below, is 20 mA, a long time after the key K is closed. What was the reading of the same ammeter, immediately after the key was closed ?



(1) zero

(2) 16 mA

(3) 25 mA

(4) $\frac{70}{3}$mA

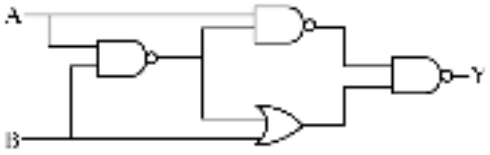
10) **Assertion :** In Zener diode the width of depletion layer is small and strength of internal electric field is high.
Reason : In Zener diode both P and N sides are heavily doped.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

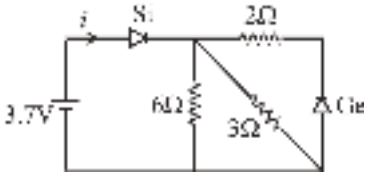
(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

11) The output of the given logic circuit is : 

(1) $\bar{A}.B$
(2) $A.\bar{B}$
(3) $A.\bar{B}+B.\bar{A}$
(4) $A.B+\bar{A}.\bar{B}$

12) Find current 'i' : 

(1) 2A
(2) 3A
(3) 1.5A
(4) 0.5A

13) In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits it is closer to the upper slit, as shown. Then, 

(1) the fringe width will increase and fringe pattern will shift down.
(2) the fringe width will remain same but fringe pattern will shift up.
(3) the fringe width will decrease and fringe pattern will shift down.
(4) the fringe width will remain same but fringe pattern will shift down.

14) An unpolarized light is falling on a liquid surface 37° with horizontal. If the reflected ray is totally polarized then refractive index of liquid will be

(1) $\frac{4}{3}$
(2) $\frac{3}{4}$
(3) $\sqrt{2}$
(4) $\sqrt{3}$

15) Read following statements carefully and select correct option.
Statement-I : Free electrons inside the metal crystel can't escape from the surface.
Statement-II : When electron attempts to escape, surface acquires positive charge.

(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are correct.

(4) Both **Statement I** and **Statement II** are incorrect.

16) If the work function of a metal is ϕ and the frequency of the incident light is υ. There is no emission of photoelectron if :-

(1) $\upsilon < \frac{\phi}{h}$

(2) $\upsilon = \frac{\phi}{h}$

(3) $\upsilon > \frac{\phi}{h}$

(4) $\upsilon \geqslant \frac{\phi}{h}$

17) In hydrogen atom transition of electron take places from second excited state to ground state, photon so produced incident on photo cathode and stopping potential found to be 8.09 volt. Then threshold frequency of the photocathode is :-

(1) 4×10^{15} Hz

(2) 10^{16} Hz

(3) 1.2×10^{10} Hz

(4) 10^{15} Hz

18) A bomb at rest explodes into two parts having masses in ratio 2 : 3 then ratio of de-Broglie wavelengths of both parts :-

(1) 1 : 2

(2) 2 : 1

(3) 2 : 3

(4) 1 : 1

19) The binding energy per nucleon of O^{16} is 7.97 MeV and that of O^{17} is 7.75 MeV. The energy (in MeV) required to remove a neutron from O^{17} is :-

(1) 3.52

(2) 3.64

(3) 4.23

(4) 7.86

20) The ground state energy of hydrogen atom is –13.6 eV, what are the kinetic and potential energies of the electron in this state :

(1) 13.6 eV, –27.2 eV

(2) 3.4 eV, –13.6 eV

(3) –13.6 eV, 27.2 eV

(4) 10.2 eV, –13.6 eV

21) A lens is made of flint glass (refractive index = 1.5). When the lens is immersed in a liquid of refractive index 1.25, the focal length

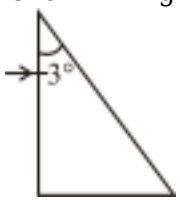
(1) increases by a factor of 1.25

(2) increases by a factor of 2.5

(3) increases by a factor of 1.2

(4) decreases by a factor of 1.2

22) Find the angle of emergence of the light ray shown in figure for following condition. The refractive index for the prism material μ = 3/2.



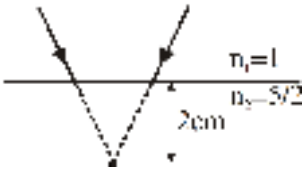
(1) 2°

(2) 3°

(3) 4.5°

(4) 1.5°

23) In the given figure rays incident on an interface would converge 2 cm below the interface if they continued to move in straight lines without bending. But due to refraction, the rays will bend and meet somewhere else. Find the distance of meeting point of refracted rays below the interface (in cm), assuming the rays to be making small angles with the normal to the interface.



(1) 5

(2) 3

(3) 2.5

(4) 1.5

24) A prism of refractive index μ and angle A is placed in minimum deviation position. If the angle of minimum deviation is equal to the angle A, then the value of A is.

(1) $\sin^{-1}\left(\frac{\mu}{2}\right)$

(2) $2\sin^{-1}\left(\sqrt{\frac{\mu}{2}}\right)$

(3) $2\sin^{-1}\left(\sqrt{\frac{1-\mu^2}{2}}\right)$

(4) $2\sin^{-1}\left(\sqrt{\frac{4-\mu^2}{4}}\right)$

25) A point object is placed on the principal axis at 60 cm in front of a concave mirror of focal length 40cm on the principal axis. If the object is moved with a velocity of 10 cm/s along the principal axis, then the velocity (only magnitude) of image is

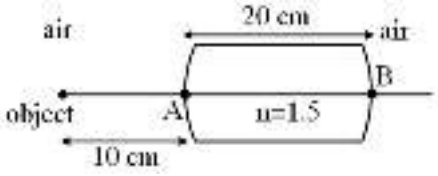
(1) 20 cm/s

(2) 30 cm/s

(3) 40 cm/s

(4) 60 cm/s

26) An object is placed 10 cm away from a glass piece (n = 1.5) of length 20 cm bound by spherical surfaces of radii of curvature 10 cm. Then the position of final image formed after two refractions at the spherical surfaces, is -



(1) 50 cm left of B

(2) 40 cm right of B

(3) 50 cm right of B

(4) 60 cm right of B

27) A parallel plate capacitor of capacitance C is connected to a battery and is charged to a potential difference V. Another capacitor of capacitance 2C is similarly charged to a potential difference 2V. The charging battery is then disconnected and the capacitors are connected in parallel to each other in such a way that the positive terminal of one is connected to the negative terminal of the other. The final energy of the configuration is :-

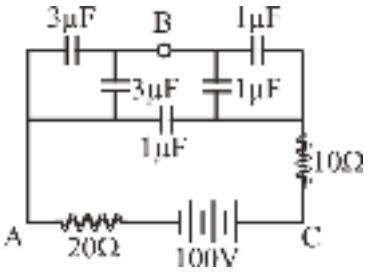
(1) Zero

(2) $\frac{3}{2}CV^2$

(3) $\frac{25}{6}CV^2$

(4) $\frac{9}{2}CV^2$

28) In the figure below, what is the potential difference between the point A and B and between B and C respectively in steady state :-



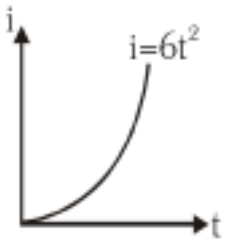
(1) V_{AB} = 100 volts, V_{BC} = 100 volts

(2) V_{AB} = 75 volts, V_{BC} = 25 volts

(3) V_{AB} = 25 volts, V_{BC} = 75 volts

(4) V_{AB} = 50 volts, V_{BC} = 50 volts

29) The current i through a wire varies with time t as shown. How many charge flows through the wire in time interval t_1 = 0 to t_2 = 3 second :-



(1) 13.5 C

(2) 27 C

(3) 37.5 C

(4) 54 C

30) Masses of 3 wires of same metal are in the ratio 1 : 2 : 3 and their lengths are in the ratio 3 : 2 : 1. The electrical resistances are in ratio :

(1) 1 : 4 : 9

(2) 9 : 4 : 1

(3) 1 : 2 : 3

(4) 27 : 6 : 1

31) A 30 V, 90 W lamp is to be operated on a 120 V DC line. For proper glow, a resistor of...Ω should be connected in series with the lamp.

(1) 40

(2) 10

(3) 20

(4) 30

32) The electric field and the electric potential at a point are E and V respectively
(A) If E = 0, V must be zero
(B) If V = 0, E must be zero
(C) If E ≠ 0, V cannot be zero
(D) If V ≠ 0, E cannot be zero

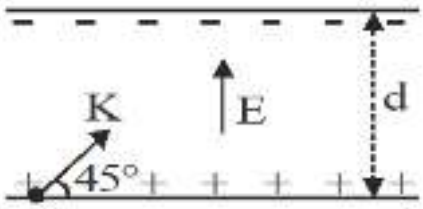
(1) A, B, C, D

(2) B, C, D

(3) B, D

(4) None

33) An electron is projected as in figure with kinetic energy K, at an angle θ = 45° between two charged plates. The magnitude of the electric field so that the electron just fails to strike the upper



plate, should be greater than:

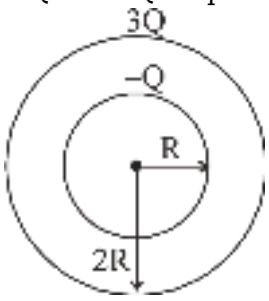
(1) $\frac{K}{ed}$

(2) $\frac{2K}{ed}$

(3) $\frac{K}{2ed}$

(4) Infinite

34) Two conducting spherical shells of radii R and 2R carry charges – Q and 3Q respectively. How much charge will flow into the earth if the inner shell is grounded ?



(1) $-\frac{Q}{2}$

(2) $\frac{Q}{2}$

(3) $\frac{3Q}{2}$

(4) $-\frac{3Q}{2}$

35) A circular loop of radius R carries a total charge Q distributed uniformly over its circumference. A small length of arc subtending angle θ at the centre is cutoff. Find out the electric field at the centre due to remaining portion.

(1) $\frac{Q}{4\pi^2\varepsilon_0 R^2}$ sin (θ)

(2) $\frac{Q}{4\pi^2\varepsilon_0 R^2}\sin\left(\frac{\theta}{2}\right)$

(3) $\frac{Q}{4\pi^2\varepsilon_0 R^2}\sin\left(\frac{\theta}{4}\right)$

(4) $\frac{Q}{4\pi^2\varepsilon_0 R^2}\sin\left(\frac{\theta}{8}\right)$

36) Between the plates of a parallel-plate capacitor, there is a metallic plate whose thickness takes up 60% of the capacitor gap. When that plate is absent the capacitor has a capacity C = 20nF. The capacitor is connected to a dc voltage source of voltage =100 V. The metallic plate is slowly extracted from gap. Find the mechanical work (in μJ) performed in the process of plate extraction.

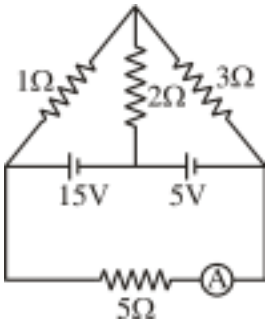
(1) 150

(2) 100

(3) 250

(4) 200

37) Reading of Ammeter will be :-



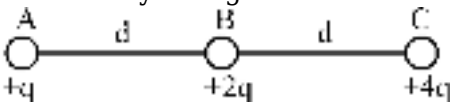
(1) 1A

(2) 2A

(3) 3A

(4) 4A

38) Three charges +q, +2q and +4q are connected by strings as shown in the figure. What is the ratio of tensions in the strings AB and BC.



(1) 1 : 2

(2) 1 : 3

(3) 4 : 1

(4) 4 : 3

39) A charged particle enters a uniform magnetic field with velocity vector at an angle of 45° with the magnetic field. The pitch of the helical path followed by the particle is p. The radius of the helix will be :-

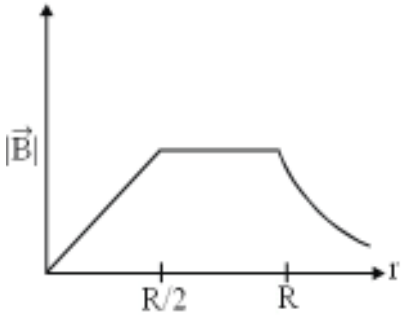
(1) $\frac{p}{\sqrt{2}\pi}$

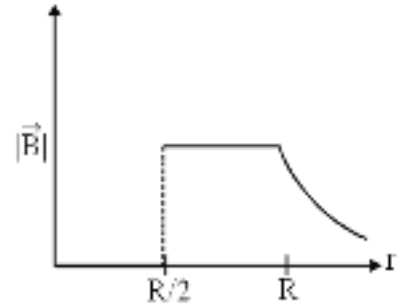
(2) $\sqrt{2}p$

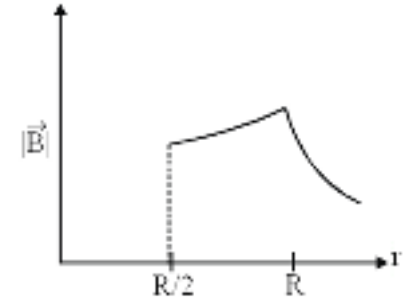
(3) $\frac{p}{2\pi}$

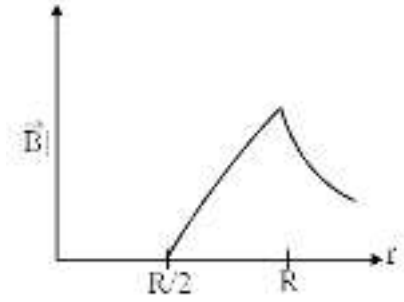
(4) $\frac{\sqrt{2}p}{\pi}$

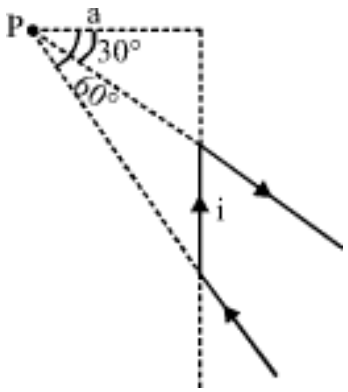
40) An infinitely long hollow conducting cylinder with inner radius R/2 and outer radius R carries a uniform current density along its length. The magnitude of the magnetic field, $|\vec{B}|$ as a function of the radial distance r from the axis is best represented by :-

(1) 

(2) 

(3) 

(4) 



41) Find magnetic field at point P in given figure :-

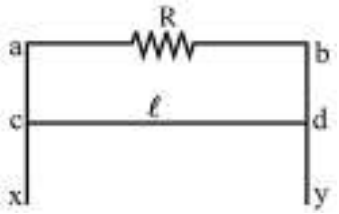
(1) $\frac{\mu_0 i}{4\pi a}\left(\sqrt{3}-1\right)$

(2) $\frac{\mu_0 i}{8\pi a}\left(1-\sqrt{3}\right)$

(3) $\frac{\mu_0 i}{8\pi a}\left(\sqrt{3}-1\right)$

(4) $\frac{\mu_0 i}{4\pi a}\left(1-\sqrt{3}\right)$

42) A wire cd of length and mass m is sliding without friction on conducting rails ax and by as shown. The vertical rails are connected to each other with a resistance R between a and b. A uniform magnetic field B is applied perpendicular to the plane abcd such that cd moves with a constant velocity of



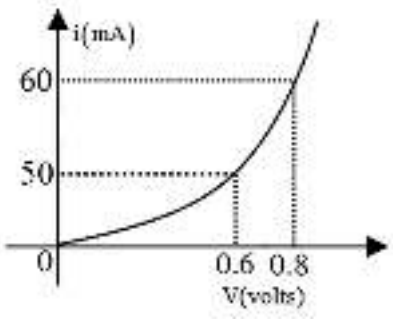
(1) $\frac{mgR}{B^2\ell^2}$

(2) $\frac{mgR}{B\ell}$

(3) $\frac{mgR}{B^3\ell^3}$

(4) $\frac{mgR}{B^2\ell}$

43) The forward characteristic of P-N junction diode is as shown in figure below, the dynamic resistance of diode at applied potential of 0.7V is



(1) 5Ω

(2) 10Ω

(3) 15Ω

(4) 20Ω

44) A travelling microscope has 20 divisions in 1 cm of its main scale. Also its 50 vernier scale divisions are equal to 48 main scale divisions. The least count of travelling microscope is given by x × 10^{-3} cm, where x is equal to :-

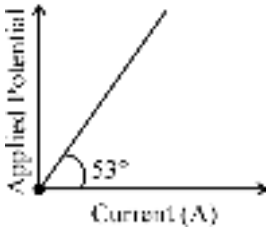
(1) 1

(2) 2

(3) 4

(4) 5

45) The variation of applied potential and current flowing through a given wire is as shown in figure. The length of the wire is 50 cm and diameter of wire is measured to be 0.2 cm. The resistivity of the wire, is given near to :



(1) 2 × 10^{-4} Ω-m

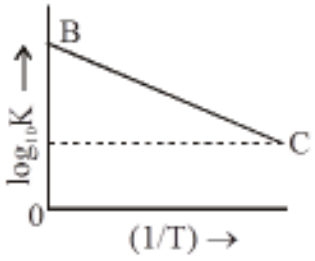
(2) 4 × 10^{-6} Ω-m

(3) 8 × 10^{-6} Ω-m

(4) 4 × 10^{-4} Ω-m

CHEMISTRY

1) Following figure shows a graph in $\log_{10}K$ vs $\frac{1}{T}$ where K is rate constant and T is temperature. The straight line BC has slope, $\tan\theta = -\frac{1}{2.303}$ and an intercept of 5 on Y–axis. Thus the energy of activation E_a is :-



(1) 2.303 × 2 cal mol^{-1}

(2) 2/2.303 cal mol^{-1}

(3) 2 cal mol^{-1}

(4) None of these

2) If rate of formation of SO_3 is 0.8 $glit^{-1}sec^{-1}$ then calculate the rate of disappearance of O_2 in $glit^{-1}sec^{-1}$ for the reaction $2SO_2 + O_2 \rightarrow 2SO_3$

(1) 0.4 $glit^{-1}sec^{-1}$

(2) 0.16 $glit^{-1}sec^{-1}$

(3) 0.8 $glit^{-1}sec^{-1}$

(4) 1.6 $glit^{-1}sec^{-1}$

3) The half-life of decomposition of gaseous CH_3CHO at initial pressure of 273 mm and 182 mm of Hg were 446 sec and 669 sec respectively. The order of the reaction is :-

(1) 0

(2) 1

(3) 2

(4) 1.5

4) The compound A decomposed (A → product) by first order kinetics. If the concentration of A dropped from 0.1 M to 0.05 M in 30 minutes, find the rate of reaction after 60 minutes, from start in $M\ min^{-1}$

(1) $\frac{\ln 2}{1200}$

(2) $\frac{\ln 2}{600}$

(3) $\frac{\ln 2}{120}$

(4) $\frac{\ln 2}{60}$

5)

Find the ratio of time $t_{1/8}$ to $t_{15/16}$ for zero order reaction.

(1) 4 : 15

(2) 2 : 15

(3) 1 : 15

(4) None of these

6) The electrochemical equivalents of two substances are E_1 and E_2. The current required to deposit the same amount at the cathodes in same time must be in ratio of :-

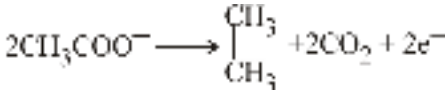
(1) $E_1 : E_2$

(2) $(E_1 - E_2) : E_2$

(3) $E_1 : (E_2 - E_1)$

(4) $E_2 : E_1$

7) Ethane gas is obtained in Kolbe's electrolysis of CH_3COONa according to the following reaction


What volume of gas (ethane gas) at S.T.P. would be obtained by a current of 0.5 amp (80% efficient); if the current is passed for 965 min ?

(1) 1.8 L

(2) 2.688 L

(3) 3.45 L

(4) 11.2 L

8) pH of the solution in the anode half cell of the following cell at 25ºC is x when $E_{cell} - E^{º}_{cell} = 0.0591$ V, Pt(H_2) (1 atm) | pH = x | | Ni^{2+} (1 M) | Ni. Find out x.

(1) 1

(2) 3

(3) 6

(4) 2

9) Standard reduction potentials of four metal electrodes are
$A^+/A = -0.250$ V , $B^+/B = -0.140$ V
$C^+/C = -0.126$ V , $D^+/D = -0.402$ V
The metal that displaces A^+ from aqueous solution of its compounds is :-

(1) B

(2) C

(3) D

(4) None of the above

10) The molar conductivity of $AgNO_3$, AgCl and NaCl at infinite dilution are 116.5, 121.6 and 110.3 S cm^2mol^{-1} respectively. The molar conductivity of $NaNO_3$ at infinite dilution is :

(1) 211.7

(2) 105.2

(3) 118.8

(4) 342.7

11) A solution of isopropyl alcohol and propyl alcohol has a vapour pressure 200 mm of Hg if it has 25% mole of isopropyl alcohol. Another solution of same components containing 25% mole propyl alcohol has vapour pressure 300 mm of Hg. Then vapour pressure of pure isopropyl alcohol in mm of Hg is:-

(1) 50

(2) 150

(3) 250

(4) 350

12) 17.4% (mass/volume) K_2SO_4 solution at 27°C is isotonic to 5.85% (mass/volume) NaCl solution at

27°C. If NaCl is 100% ionised, what is % ionisation of K_2SO_4 in aq. solution ?
[At wt. of K = 39, Na = 23]

(1) 25

(2) 50

(3) 75

(4) 100

13) One molal solution of a carboxylic acid in benzene shows the elevation of boiling point of 1.518 K. The percentage of association for dimerization of the acid in benzene is (K_b for benzene = 2.53 K kg mol^{-1}) :-

(1) 80%

(2) 50%

(3) 75%

(4) 100%

14) The osmotic pressure of 2.22 % (w/v) $CaCl_2$ solution at 27ºC is
(R = 0.08 L-atm/K-mol)
[Ca = 40, Cl = 35.5]

(1) 14.4 torr

(2) 3.42 atm

(3) 3.42 torr

(4) 14.4 atm

15) Consider 1 M solution of X and Y respectively. If X undergoes 80% dimerisation whereas Y undergoes 90% trimerisation, then which of the following statement is correct :-

(1) B.P. of X < B.P. of Y

(2) Osmotic pressure of Y > Osmotic pressure of X

(3) F.P. of X < F.P. of Y

(4) Colligative property of X = Colligative property of Y

16) Which of the following compounds of nitrogen is coloured ?

(1) NO_2

(2) NH_3

(3) N_2O

(4) N_2O_4

17) Which of the following is covalent carbide ?

(1) CaC_2

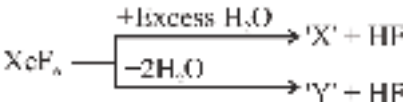
(2) Al_4C_3

(3) SiC

(4) Be_2C

18) Which of the following act as an oxidising as well as a reducing agent ?

(1) HNO_2
(2) H_2O_2
(3) SO_2
(4) All of these

19) Incorrect statement regarding following reactions is : 

(1) 'X' is explosive
(2) 'Y' is an oxyacid of xenon
(3) Both are example of non-redox reaction
(4) XeF_6 can undergo partial hydrolysis

20)

Which of the following compound is used as the starting material for the preparation of potassium dichromate :-

(1) $K_2SO_4.Cr_2(SO_4)_3.24H_2O$ (Chrome alum)
(2) $PbCrO_4$ (Chrome yellow)
(3) $FeCr_2O_4$ (Chromite)
(4) $PbCrO_4PbO$ (Chrome red)

21) Which of the following statements is false :-

(1) $Cr_2O_7^{2-}$ has Cr – O – Cr bond
(2) CrO_4^{2-} is tetrahedral in shape
(3) $Na_2Cr_2O_7$ is a primary standard in volumetry analysis
(4) $Na_2Cr_2O_7$ is more soluble than $K_2Cr_2O_7$

22) Which of the following statements containing transition elements is not true ?

(1) They are all metals
(2) They easily form complexes
(3) Compounds containing their ions are coloured
(4) They show multiple oxidation states which always differing by two units

23) $(NH_4)_2Cr_2O_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is :-

(1) Cr_2O_3

(2) CrO_2

(3) Cr_2O_4

(4) CrO_3

24) $[Ni(CO)_4]$ and $[Ni(NH_3)_4]^{2+}$ do not differ in

(1) magnetic moment

(2) oxidation number of Ni

(3) geometry

(4) EAN

25) Geometry and hybridisation of complex $[Cu(NH_3)_4]^{2+}$:

(1) Square planar → dsp^2

(2) Tetrahedral → sp^3

(3) Octahedral → sp^3d^2

(4) Octahedral → d^2sp^3

26) The value of n in the carbonyl is
$(CO)_n$ — Co – Co — $(CO)_n$

(1) 4

(2) 5

(3) 6

(4) 8

27) Total number of geometrical isomers for square planar shape complex ion $[M(A)(B)(C)(D)]^{+x}$ where, A, B, C and D are monodentate ligand.

(1) 2

(2) 3

(3) 4

(4) 6

28) An unknown compound gives positive chromyl chloride test. That compound forms blue solution with potassium ferrocyanide. Predict that compound :-

(1) NaCl

(2) $FeBr_3$

(3) $CuCl_2$

(4) $FeCl_3$

29) The acidic solution of a salt produced a deep blue colour with starch iodide solution. The salt may be

(1) chloride

(2) nitrite

(3) acetate

(4) Carbonate

30) $K_4[Fe(CN)_6]$ can be used to detect one or more out of Fe^{+2}, Fe^{+3}, Zn^{+2}, Cu^{+2}

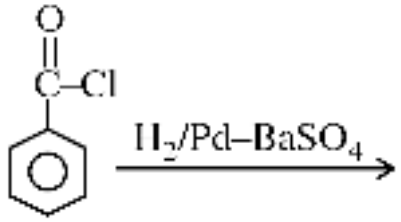
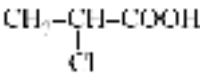
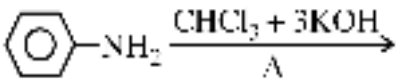
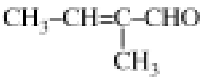
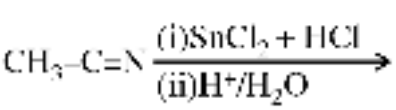
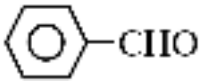
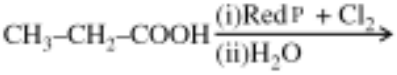
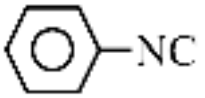
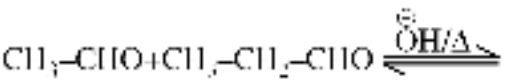
(1) Only Fe^{+2}, Fe^{+3}

(2) Only Fe^{+3}, Zn^{+2}, Cu^{+2}

(3) All but not Fe^{+3}

(4) All of these

31) Match the column

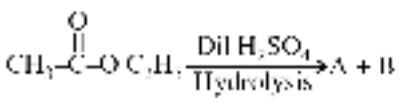
Column-I (Reaction)		**Column-II (Product & Name of reaction)**	
(a)		(i)	 HVZ Reaction.
(b)		(ii)	 Aldol condensation
(c)		(iii)	 Rosenmund Reduction
(d)		(iv)	 Carbylamine Reaction
(e)		(v)	CH_3–CHO Stephen Reduction

(1) a→(iii), b→(iv), c→(v), d→(i), e→(ii)

(2) a→(iii), b→(iv), c→(i), d→(v), e→(ii)

(3) a→(v), b→(iv), c→(iii), d→(i), e→(ii)

(4) a→(iv), b→(iii), c→(i), d→(v), e→(ii)

32) 

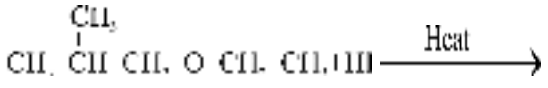
Correct statement for product A & B in above reaction?

(1) Both Product A & B react with Na, NaOH, $NaHCO_3$

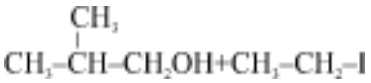
(2) Both Product A & B gives H.V.Z reaction with Red P + Cl_2

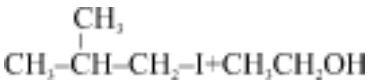
(3) One Product gives $CHCl_3$ on reaction with Cl_2 + NaOH

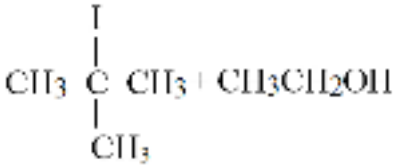
(4) Both Product gives Aldehyde on react with Cu/300°C

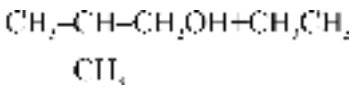
33) In the reaction  .....

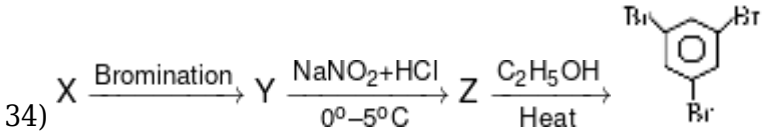
Which of the following compounds will be formed as major product ?

(1) 

(2) 

(3) 

(4) 

34) 

Correct statement For X, Y, Z are ?

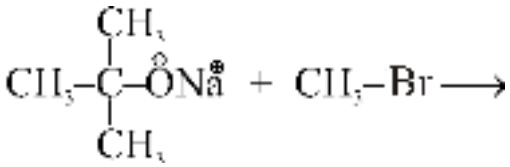
(1) X react with Hinsberg's reagent and soluble in KOH

(2) Y is ortho & para mono Bromo product of X

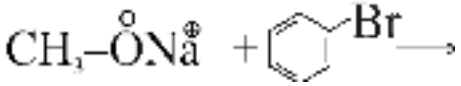
(3) Z gives phenol on reaction with H_2O/boil

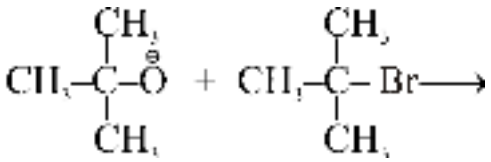
(4) X gives phenol on reaction with HNO_2 / 0°–5°C

35) Which of the following reaction will produce ether as major product?

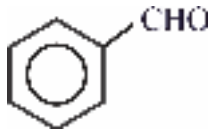
(1) 

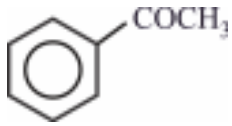
(2) 

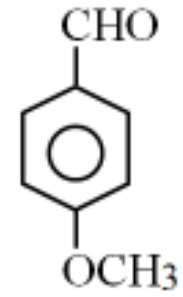
(3) 

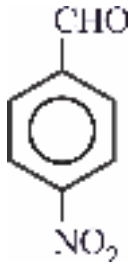
(4) 

36) Which one is most reactive towards nucleophilic addition reaction?

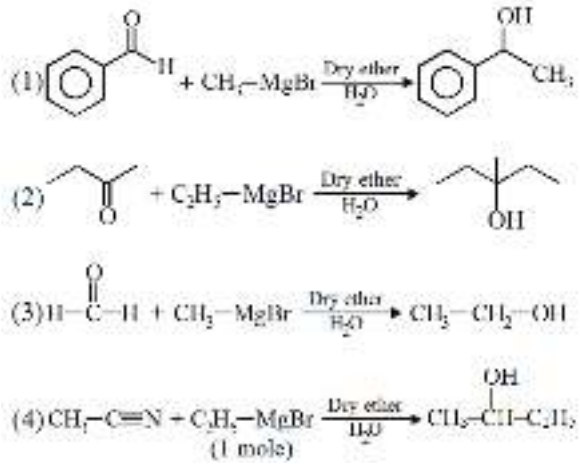
(1) 

(2) 

(3) 

(4) 

37) Find out the reactions which show correct product.



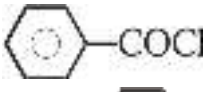
(1) 1, 2, 3, 4

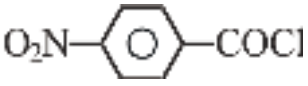
(2) Only 1, 2, 3

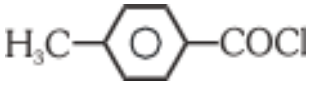
(3) Only 1, 2

(4) Only 3, 4

38) Consider the following compounds:

(i) 

(ii) 

(iii) 

(iv) 

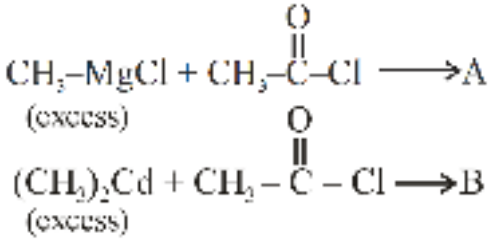
The correct order of reactivity towards hydrolysis is:-

(1) (i) > (ii) > (iii) > (iv)

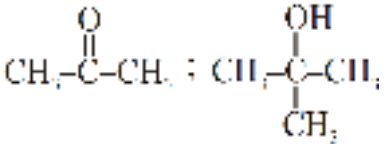
(2) (iv) > (ii) > (i) > (iii)

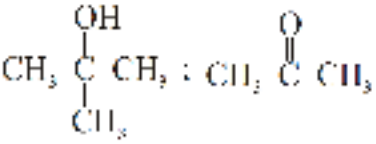
(3) (ii) > (iv) > (i) > (iii)

(4) (ii) > (iv) > (iii) > (i)

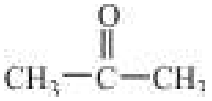


39) In the following reaction : Product A and B are respectively :

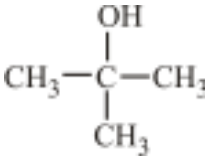
(1) 

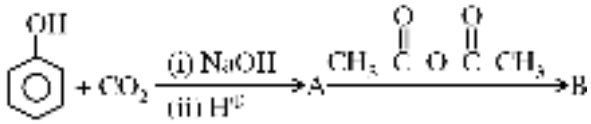
(2) 

Both is

(3) 

Both is

(4) 

40) 

Incorrect statement for product A & B are :-

(1) A → Salicylic Acid B → Aspirin

(2) A & B gives voilet colour with neutral $FeCl_3$

(3) A & B both reacts with $NaHCO_3$

(4) A & B both contain carboxylic group

41) The correct set of the products obtained in the following reactions – (A) RCN $\xrightarrow{\text{reduction}}$

(B) RCN $\xrightarrow[\text{(ii) } H_2O]{\text{(i) } CH_3MgBr}$

(C) RNC $\xrightarrow{\text{hydrolysis}}$

(D) $RNH_2 \xrightarrow{HNO_2}$ The correct answer is -

(1) A - 2° Amine; B - Methyl ketone; C - 1° Amine; D - Alcohol

(2) A - 1° Amine; B- Methyl ketone; C - 1° Amine; D - Alcohol

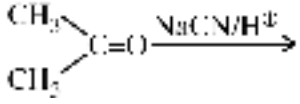
(3) A - 2° Amine; B - Methyl ketone; C - 2° Amine; D - Acid

(4) A - 2° Amine; B - Methyl ketone; C -2° Amine; D - Aldehyde

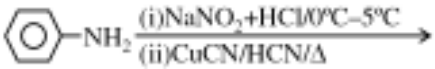
42) Consider the following method :

(I) $CH_3-CH_2-Br \xrightarrow{NaCN}$

(II) $CH_3-CH_2-Cl \xrightarrow{AgCN}$

(III) 

(IV) $CH_3-CH_2-OH \xrightarrow{KCN}$

(V) 

Which of following method can be used to prepare cyanide compound :-

(1) Only I, II, III, IV

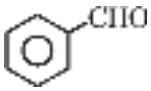
(2) Only II, III, IV

(3) Only I, III, V

(4) Only I, III, IV, V

43)

Which is wrongly match

	Pair	Distinguish Reagent
1	Methyl Alcohol and Ethyl Alcohol	I_2 + Base
2	 and CH_3–CHO	Fehling Solution
3	Phenol and Carboxylic Acid	$NaHCO_3$
4	Propane-1-Amine and propane-2-Amine	Hinsberg reagent

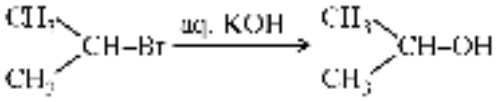
(1) 1

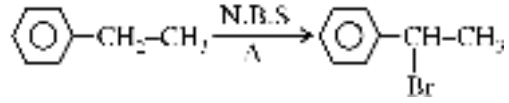
(2) 2

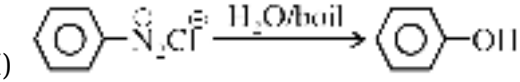
(3) 3

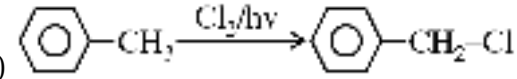
(4) 4

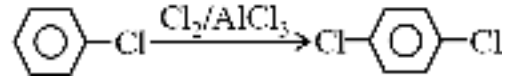
44) For following reaction :

(I) 

(II) 

(III) 

(IV) 

(V) 

Which of following statement is correct

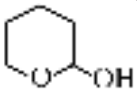
(1) Reaction – I, III – Nucleophilic substitution reaction
Reaction – II, IV – Free radical substitution reaction
Reaction – V – Electrophilic substitution reaction

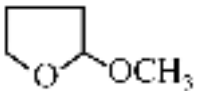
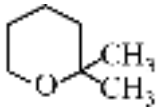
(2) Reaction – II, III – Nucleophilic substitution reaction
Reaction – IV, V – Free radical substitution reaction
Reaction – III – Electrophilic substitution reaction

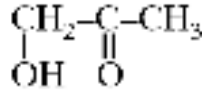
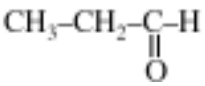
(3) Reaction – I, II, III – Nucleophilic substitution reaction
Reaction – IV, V – Free radical substitution reaction

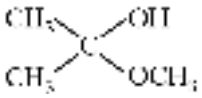
(4) Reaction – I, III – Nucleophilic substitution reaction
Reaction – II, IV, V – Free radical substitution reaction

45) How many compound will give silver mirror with Tollen's reagent -

(I)  (II) Ph–NH–OH

(III)  (IV) 

(V)  (VI) 

(VII)  (VIII) 

(1) 5

(2) 6

(3) 4

(4) 3

BIOLOGY

1) In pinnately compound leaf, a number of leaf lets are present on a common axis called.

(1) Petiole

(2) Branch

(3) Rachis

(4) Midrib

2) Select the correct option regarding the following statement-
a. Polyarch condition
b. Pith is large and well developed
c. Do not undergo any secondary growth.

(1) Dicot root

(2) Monocot root

(3) Dicot stem

(4) Monocot stem

3) Identify the correct match from the columns I, II and III–

Column-I	Column-II	Column-III
(1) Alternate phyllotaxy	(a) The leaflets are attached at a common point i.e.; at the tip of petiole.	(i) *Alstonia*
(2) Opposite phyllotaxy	(b) A pair of leaves arise at each node	(ii) Silk cotton
(3) Whorled phyllotaxy	(c) More than two leaves arise at each node	(iii) China rose
(4) Palmately compound leaf	(d) A single leaf arises at each node	(iv) Guava

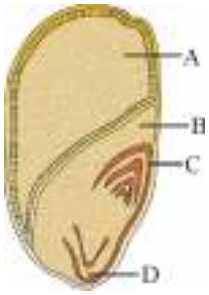
(1) 2–a–iii, 1–c–iv, 3–b–i, 4–d–ii

(2) 1–c–iv, 2–d–iii, 3–b–i, 4–a–ii

(3) 1–d–iii, 2–b–iv, 3–c–i, 4–a–ii

(4) 2–a–iii, 1–c–iv, 3–b–ii, 4–d–i

4) Given below is the diagram of monocotyledons seed identify the parts labelled A, B, C, D.



	Part A	Part B	Part C	Part D
(1)	Endosperm	Scutellum	Coleoptile	coleorhiza
(2)	Scutellum	Endosperm	coleorhiza	Coleoptile
(3)	Scutellum	coleorhiza	Coleoptile	Endosperm
(4)	coleorhiza	Coleoptile	Scutellum	Endosperm

(1) 1

(2) 2

(3) 3

(4) 4

5) Given below are two Statements.
Statements-I : The shape, margin, Apex, surface and extent of incision of lamina varies in different leaves.
Statements-II : veins provide rigidity to the leaf blade and act as channels of transport for water, minerals and food materials.
Choose the correct answer from the option given below.

(1) Both Statement-I and II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and II are correct

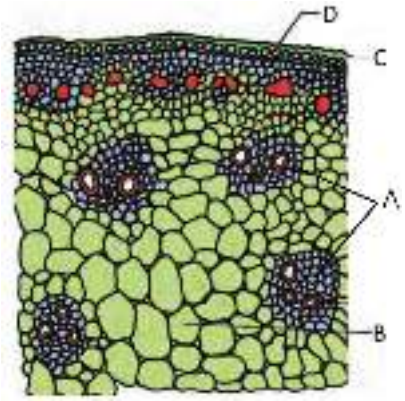
6) Among Chinarose, Mustard, Brinjal, Potato, Guava, Cucumber, Onion and tulip.
How many plants have superior ovary?

(1) Three

(2) Five

(3) Six

(4) Four



7) Observe the given diagram and choose the correct labeling

	A	B	C	D
(1)	Vascular bundle	Ground tissue	Epidermis	Hypodermis
(2)	Vascular bundle	Phloem	Xylem	Ground tissue
(3)	Vascular bundle	Ground tissue	Hypodermis	Epidermis
(4)	Vascular bundle	Ground tissue	Phloem	Epidermis

(1) 1

(2) 2

(3) 3

(4) 4

8) Select the incorrect statement in the followings :

(1) Open conjoint, collateral or bicollateral vascular bundles found in monocot stem

(2) All the tissues on the innerside of the endodermis such as pericycle, vascular bundles and pith in dicot root consitute the stele

(3) Polyarch vascular bundles found in monocot root

(4) Phloem parenchyma is absent in monocot stem

9) **Assertion (A) :** Geitonogamy is just ecologically self-pollination
Reason (R) : From genetists point of view geitonogamy is equivalent to self-pollination

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

10) Which is/are incorrect
(a) Microsporangium is near circular in outline
(b) Four wall layers of microsporangium help in dehiscence of anther
(c) Vegetative cell nucleus of pollen grain is irregular in shape
(d) Pollen grains are about 25-50 Å in diameter

(1) only a

(2) only b, c

(3) All

(4) only b, d

11) Which is incorrect :

(1) Sporogenous tissue undergo meiotic division to form microspore tetrad.

(2) A typical angiosperm embryosac at maturity 8 nucleate is 7 celled.

(3) Method of embryo sac formation from a single megaspore is monosporic development.

(4) Wind pollination requires heavy and sticky pollen grains for easy transportation.

12) Match the options

	Structure		**Shape**
(a)	Vegetative nucleus	(i)	Spindle shaped
(b)	Microsporangia	(ii)	Spherical shaped
(c)	Pollen grain	(iii)	Large irregular shape
(d)	Generative cell	(iv)	Near circular in outline

(1) a-iv, b-iii, c-i, d-ii

(2) a-iii, b-iv, c-ii, d-i

(3) a-i, b-ii, c-iii, d-iv

(4) a-ii, b-i, c-iv, d-iii

13) Wind pollinated plants generally do not show the character :

(1) Feathery stigmas

(2) Single ovule in the ovary

(3) Well exposed stamens

(4) Flowers are large and colourful

14) **Statement I :** Cleistogamy do not favours inbreeding depression.
Statement II : Cleistogamy is disadvantageus in respect to no further development of variation in offsprings

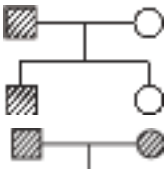
(1) Statement I & II are correct

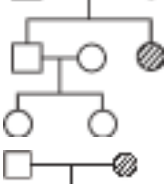
(2) Statement I & II are incorrect

(3) Statement I is correct, II is incorrect

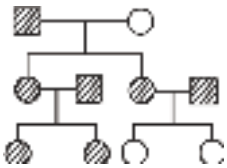
(4) Statement I is incorrect, II is correct

15) Which of the following pedigree can be sex-linked haemophilia ?

(1) 

(2) 

(3) 

(4) 

16) Study the two cases carefully :

	Mother	Father	Children
Case I	With disease	Normal	Sons always disese
Case II	Normal	with disease	Son and daugthers could show disease

The correct interpretation of the two cases is :-

(1) Case I - X-Linked recessive disease
Case II - Autosomal recessive disease

(2) Case I - X-Linked recessive disease
Case II - X-Linked dominant disease

(3) Case I - Y-Linked disease
Case II - X-Linked recessive disease

(4) Case I - Y-Linked
Case II - Autosomal recessive disease

17) In a family, a father with normal color vision and a mother who is a carrier for color blindness have a son. What is the probability that the son will have color blindness?

(1) 0%

(2) 25%

(3) 50%

(4) 75%

18) **Assertion:** Female heterogamety is observed in birds.
Reason: Female birds produce two different types of gametes in terms of sex chromosomes, designated as Z and W chromosomes.

(1) Both assertion and reason are true, and the reason is the correct explanation of the assertion.

(2) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.

(3) The assertion is true, but the reason is false

(4) The assertion is false.

19) How many of the given are autosomal recessive disorder ?
Haemophilia, Cystic fibrosis, Phenyl ketonuria, Thalassemia, Sickle cell anaemia, Myotonic-dystrophy.

(1) 4

(2) 5

(3) 3

(4) 2

20) The ratio between carrier, disease free and diseased individual on marriage between two carriers of sickle cell anaemia will be :-

(1) 0 : 2 : 1

(2) 0 : 1 : 1

(3) 1 : 2 : 1

(4) 2 : 1 : 1

21) Read the following statements and find out incorrect one :-
(I) Flower colour in *Mirabilis* shows complete dominance.
(II) Coat colour in cattles shows codominance.
(III) Starch grain size in garden pea shows incomplete dominance.
(IV) Wrinkled seed shape in garden pea is recessive.

(1) Only I

(2) Only II

(3) Only III

(4) I and IV both

22) Given below are two statements :
Statement-I :- The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleoid.
Statement-II :- The beads-on-string structure in chromatin is packaged to form chromatin fibers that are further coiled and condensed at metaphase stage of cell division to form chromosomes.

(1) Both statement-I and statement-II are incorrect

(2) Statement-I is correct but statement-II is incorrect

(3) Statement-I is incorrect but statement-II is correct

(4) Both statement-I and statement-II are correct

23) Which one of the following statements about histones is **wrong**?

(1) Histones are organized to form a unit of 8 molecules, known as histone octamer.

(2) The pH of histones is slightly acidic.

(3) Histones are rich in amino acids - Lysine and Arginine.

(4) Histones carry positive charge in the side chain.

24) Ten *E.coli* cells with ^{15}N - dsDNA are incubated in medium containing ^{14}N nucleotide. After 60 minutes, how many *E.coli* cells will have DNA totally free from ^{15}N ?

(1) 40 cells

(2) 60 cells

(3) 80 cells

(4) 20 cells

25) If a DNA molecule is shortened by 25 base pairs, how many helical turns will be reduced from its structure?

(1) 3

(2) 2.5

(3) 2

(4) 1

26) **Statement-I :** Severo Ochoa enzyme was used for enzymatic synthesis of RNA.
Statement-II : Hargobind khorana developed the chemical method for synthesizing RNA molecule.
Select the correct answer from the options given below :

(1) Both statement-I and statement-II are correct.

(2) Only statement-I is correct.

(3) Only statement-II is correct.

(4) Both statement-I and statement-II are incorrect.

27) **Statement-I :** Single DNA-dependent RNA polymerase catalyses transcription of all type of RNA_s in bacteria.
Statement-II : In bacteria there are three major types of RNA_s: mRNA, t-RNA and rRNA.
Select the correct answer from the options given below :-

(1) Both statement-I and statement-II are correct.

(2) Only statement-I is correct.

(3) Only statement-II is correct.

(4) Both statement-I and statement-II are incorrect.

28) Match the list A, B

	List-A		**List-B**
(A)	N-glycosidic bond	(I)	Dinucleotides
(B)	Phosphoester Bond	(II)	UAA
(C)	Stop codon	(III)	Nucleoside + Phosphate group
(D)	Phosphodiester bond	(IV)	Pentose sugar + N-base
		(V)	UAG

(1) A-V, B-IV, C-III, II, D-I

(2) A-IV, B-III, C-II,V, D-I

(3) A-I, B-III, C-II,V, D-IV

(4) A-I, B-II,III, C-V, D-IV

29) **Assertion :** Eukaryotic hn-RNA requires post transcriptional processing to produce functional m-RNA.
Reason : Eukaryotic primary transcripts contain non-functional fragments called intron.

(1) Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

30) Which of the following statement is correct ?

(1) DNA can directly involved in protein synthesis.

(2) DNA is the first to evolve and RNA was derived from DNA.

(3) In bacteria, the transcribed mRNA is nonfunctional, hence cannot be directly be translated.

(4) Regulation of transcription is the primary step for regulation of gene expression.

31) Which of the following factors affects both GPP and NPP is an ecosystem.

(1) Animal respiration

(2) Decomposition rates

(3) Availability of nitrogen and sugar to decomposer

(4) Amount of sunlight and temperature

32) Which of the following statements are correct?
A. Primary production is the amount of biomass produced per unit area over a time period by plants.
B. Gross primary productivity (GPP) is the total production of organic matter during photosynthesis.
C. Net primary productivity (NPP) is the GPP minus respiration losses.
D. Secondary productivity refers to the rate of formation of new organic matter by decomposers.
E. The productivity of oceans is low despite covering 70% of the Earth's surface.
Choose the correct answer:

(1) A, B, C only

(2) A, B, C, E only

(3) B, C, D only

(4) A, D, E only

33) Match **List–I** with **List–II** :

	List–I		**List–II**
A.	First trophic level	I.	Birds, Wolf
B.	Second trophic level	II.	Phytoplanktons, Grass
C.	Third trophic level	III.	Zooplanktons, Cow
D.	Forth trophic level	IV.	Man, Lion

Choose the **correct** answer from the options given below :

(1) A-II, B-IV, C-III, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-II, B-III, C-I, D-IV

(4) A-II, B-IV, C-I, D-III

34) Why are relationships such as food chains, food webs, and nutrient cycles important in an ecosystem ?

(1) They help to maintain the balance of energy and nutrients within the system.

(2) They prevent the loss of energy in all organisms.

(3) They help organisms to avoid interactions with their physical environment.

(4) They allow only large ecosystem like forest to function properly.

35) Which of the following statement is incorrect ?

(1) Vertical arrangement of different species occupying different levels is called stratification.

(2) Net primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.

(3) Bacterial and fungal enzymes degrade detritus into simpler inorganic substances this process is called as catabolism.

(4) Each trophic level has a certain mass of living material at a particular time called as the standing crop.

36) Which of the following statements are correct?
A. Competition affects the fitness of competing species.
B. In mutualism, plants offer nectar and pollen to pollinators as rewards.
C. Resource partitioning allows co-existing species to avoid competition.
D. Parasites evolve adaptations to counteract host defences.
E. Mycorrhizae are mutualistic associations between fungi and plant roots.
Choose the correct answer:

(1) A, B, C only
(2) A, B, C, D, E All
(3) A, B, D only
(4) A, C, E only

37) Match **List–I** with **List–II** :

	List–I		**List–II**
A.	Predation	I.	Orchid on mango tree
B.	Mutualism	II.	Sparrow eating a seed
C.	Commensalism	III.	Fig and wasp
D.	Competition	IV.	Goat and Abington tortoise

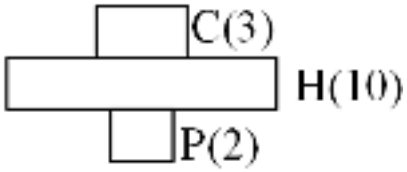
Choose the **correct** answer from the options given below :
(1) A-II, B-III, C-I, D-IV
(2) A-I, B-II, C-III, D-IV
(3) A-IV, B-II, C-III, D-I
(4) A-II, B-IV, C-I, D-III

38) Which statement about competitive release is true?

(1) It occurs when a species expands its range after a competitor is removed.
(2) It refers to the extinction of weaker competitors.
(3) It is only observed in controlled laboratory experiments.
(4) It occurs when competition intensifies due to resource scarcity.

39) Resource partitioning is required for :-
(a) Competitive exclusion
(b) Reduce competition
(c) Utilisation of same resources in same habitat
(d) to maintain ecological balance

(1) Only a & c
(2) Only b & d
(3) Only a & d
(4) b, c, d



40)
above pyramid represents :-

(1) Pyramid of number, in a grassland

(2) Pyramid of biomass, in lake

(3) Pyramid of biomass, in ocean

(4) Pyramid of biomass in tree ecosystem

41) Match the Column with respect to equation log S = log C + Z log A

	Column-I		**Column-II**
A.	S	I.	Y-intercept
B.	A	II.	Area
C.	Z	III.	Species richness
D.	C	IV.	Slope of the line (regression coefficient)

Choose the correct answer from the options given below :

(1) A-II, B-III, C-I, D-IV

(2) A-I, B-II, C-III, D-IV

(3) A-III, B-II, C-IV, D-I

(4) A-II, B-IV, C-I, D-III

42) Two statements are given below :
Statement-I : Sacred groves - Western Ghats and Sri Lanka, Indo-Burma and Himalaya - cover our country's exceptionally high biodiversity regions.
Statement-II : Hotspots are found in Khasi and Jaintia Hills in Meghalaya, Aravalli Hills of Rajasthan, Western Ghat regions of Karnataka and Maharashtra and the Sarguja, Chanda and Bastar area of Madhya Pradesh.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

43) **Assertion (A) :** The Nile perch introduced into lake Victoria in East Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.
Reason (R) : When alien species are introduced unintentionally or deliberately for whatever purpose, some of them turn invasive, and cause decline or extinction of indigenous species.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct and Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

44) Which of the following statements is not correct?

(1) The colonisation of tropical Pacific Islands by humans is said to have led to the extinction of more than 2,000 species of native birds.

(2) The IUCN Red List (2004) documents the extinction of 784 species (including 338 vertebrates, 359 invertebrates and 87 plants) in the last 500 years.

(3) Over-exploitation is the most important cause driving animals and plants to extinction.

(4) Many species extinction in the last 500 years (Steller's sear cow, passenger pigeon) were due to overexploitation by humans.

45) **Statement-I :** Presently, 12 percent of all bird species, 23 percent of all mammal species, 32 percent of all amphibian species and 31 percent of all gymnosperm species in the world face the threat of extinction.
Statement-II : The recent illegal introduction of the African catfish *Clarias gariepinus* for aquaculture purpose is posing a threat to the indigenous catfishes in our rivers.

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

46) When the isolation of genetic material is done the RNA can be removed by treatment with :

(1) Protease

(2) Chitinase

(3) Ribonuclease

(4) Deoxyribonuclease

47) In gel electrophoresis, separated bands of DNA are cut out from the agarose gel and extracted from the gel pieces, This step is known as :

(1) Blotting

(2) Elution

(3) Cloning

(4) Tagging

48) Choose the incorrect match :

(1) Ori - Responsible for controlling the copy number of linked DNA.

(2) Rop - Codes for the protein involved in replication of plasmid.

(3) Hind III - Restriction site present on pBR322

(4) Sal I - Selectable marker on plasmid of *Salmonella typhimurium* .

49)

Match the Column-I and II and select correct option-

Column-I		Column-II	
(a)	Selectable marker	(i)	Plasmid
(b)	Restriction enzyme	(ii)	Code for protein involved in DNA replication
(c)	rop	(iii)	Chloramphenicol resistance gene
(d)	pBR322	(iv)	Cla I

(1) (a–iv); (b–iii); (c–i); (d–ii)

(2) (a–iii); (b–iv); (c–ii); (d–i)

(3) (a–iii); (b–iv); (c–i); (d–ii)

(4) (a–iii); (b–i); (c–ii); (d–i)

50) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion : Polymerase chain reaction is employed to check the progression of a restriction enzyme digestion.
Reason (R): Restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzyme, at the optimal conditions for that specific enzyme.
Choose the correct answer from the options given below:

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

51) Which of the following enzyme(s) are essential for gene cloning?
(A) Restriction enzymes
(B) DNA recombinase
(C) Ribozyme
(D) Polymerase enzymes
(E) Ligases
Choose the correct answer from the options given below:

(1) B, D and E only

(2) A, B and E only

(3) A, D and E only

(4) D and E only

52) While studying a cloning experiment, a researcher observed that after inserting a foreign DNA into a plasmid, the transformed bacterial colonies were able to grow on ampicilin containing medium but failed to grow on tetracyclin containing medium. The plasmid was also found to contain rop gene.

Based on this observation what is the most appropriate conclusions ?

(1) Foreign DNA is inserted at the amp^R gene.

(2) Foreign DNA is inserted at the tet'R gene.

(3) The bacteria are non-transformed

(4) The plasmid lacks origin of replication

53) Which of the following are **correct** regarding the fact that genetically modified plants are useful in many ways ?
(A) Genetic modification has decreased efficiency of mineral usage by plants so that early exhaustion of fertility of soil can be prevented.
(B) Genetic modification made crops more tolerant to cold, drought, salt, heat.
(C) Genetic modification has reduced reliance on chemical pesticides.
(D) Genetic modification has helped to reduce post harvest losses.
(E) Genetic modification has enchanced nutritional value of food.
Choose the **correct** answer from the options given below.

(1) A, B, C, D and E

(2) B, C, D and E

(3) A, B, D and E

(4) A, C and D

54) Which of the following is incorrect ?

(1) There are an estimated 200,000 varieties of rice in India alone.

(2) In 1997 an American Company got patent rights on Basmati rice through US patent and trade mark office.

(3) Gene isolated from normal bone marrow cells producing ADA is introduced in to cells at early embryonic stage is temporary cure for SCID.

(4) PCR is used to detect HIV in suspected AIDS patient.

55) **Assertion :-** Genetic engineering approval committee has been set up by Indian government.
Reason :- Introduction of GMO could have unpredictable result in the ecosystem.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

56) Find out the correct match from the column I and II.

Column - I		**Column - II**	
(a)	Bt-cotton	(i)	Chemical safety testing
(b)	RNA interference	(ii)	insect resistant
(c)	Transgenic animals	(iii)	Molecular diagnosis
(d)	PCR	(iv)	Nematode resistant plant

(1) a – ii, b – iv, c – i, d – iii

(2) a – iv, b – ii, c – iv, d – i

(3) a – iii, b – iv, c – iii, d – ii

(4) a – i, b – ii, c – iv, d – iii

57)

An American company got patent right on basmati rice, this new variety had been derived from -

(1) US varieties crossed with semi-dwarf variety

(2) Indian varieties crossed with tall variety

(3) US varieties crossed with tall variety

(4) Indian varieties crossed with semi-dwarf variety.

58) Why early diagnosis of a disease is important ?

(1) It reduces treatment costs.

(2) For effective treatment of disease.

(3) It eliminates the need for medication.

(4) It prevents all complications.

59) The amount of oxygen required by the microbes in the decomposition of organic matter is called.

(1) Chemical oxygen demand

(2) Biochemical oxygen demand

(3) Total oxygen demand

(4) Dissolve oxygen

60) Given below are two statements :
Statement-I : The large hole in 'Swiss cheese' are due to production of a large amount of CO_2 by a bacterium named *Propionibacterium sharmanii*.
Statement-II : Antibiotics are chemical substances, which help in growth of other disease causing microbes.
Choose the correct answer from the options given below :

(1) Statement-I is incorrect but Statement-II is correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Both Statement-I and Statement-II are incorrect.

(4) Both Statement-I and Statement-II are correct.

61) **Statement-I :-** Fleming, chain and Florey were awarded the nobel prize in 1945 for discovery of penicillin.
Statement-II :- The biogas production technology was developed in India by IARI and KVIC.

(1) Statement-I is correct, but statement-II is incorrect.

(2) Statement-I is incorrect, but statement-II is correct.

(3) Statement-I and II both are correct.

(4) Statement-I and II both are incorrect.

62) Read the following statements carefully and select the correct option.
A. Statins reduce cholesterol levels by promoting its breakdown in the liver.
B. Statins are fungal derivatives that block the cholesterol biosynthesis pathway.
C. *Monascus purpuerus* naturally produces statins used in medicine.
D. Statins reduce cholesterol and help lower heart disease risk.
Options :

(1) Only statements B and D are correct.

(2) Only statements A, B and D are correct.

(3) Only statements B, C and D are correct.

(4) All statements A, B, C and D are correct.

63) Select the hormone which is not correctly matched with its function :
(A) Estrogen – Endometrial hyperplasia
(B) hCG – maintains corpus luteum.
(C) Oxytocin – strong uterine contractions
(D) Prolactin – milk synthesis
(E) Progesterone – promote foetal ejection reflex

(1) A and E

(2) Only D

(3) D and E

(4) Only E

64) Read the following statements A-D.
(A) Vigorous contractions of the uterus at the end of the pregnancy cause delivery of the foetus.
(B) Parturition is induced by a complex neuroendocrine mechanism.
(C) The signals for parturition originated only from fully developed foetus.
(D) Breast-Feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby.
How many of the above statements are correct ?

(1) Two

(2) Four

(3) One

(4) Three

65) A new drug has been developed by scientists to block FSH receptors. During animal trials, the drug was injected into a female monkey. What are the expected effects of the drug on the female monkey?

(1) Multiple ova are released from the ovary

(2) Ovulation will not occur

(3) Low concentration of estrogen in blood

(4) Both (2) & (3)

66) **Assertion :** Testes are located inside the abdominal cavity in scrotum.
Reason : Higher temperature in scrotum is essential for spermatogenesis.

(1) Both Assertion and Reason are correct and Reason is correct Explanation of Assertion.
(2) Both assertion and Reason are correct but Reason is not correct explanation of Assertion.
(3) Assertion is correct, Reason is wrong.
(4) Both Assertion and Reason are wrong.

67) **Statement-I :-** The trophoblast layer gets attached to the endometrium and the inner cell mass gets differentiated as the embryo.
Statement-II :- The morula becomes embedded in the endometrium of the uterus.

(1) Both statement I and statement II are correct
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct
(4) Both statement I and statement II are incorrect

68) Match column-I with column-II and select the **correct** option using the codes given below :

Column-I		**Column-II**	
a.	Labia majora	(i)	Embryo formation
b.	Secondary oocyte	(ii)	Sperm
c.	Inner cell mass	(iii)	Female external genitalia
d.	Acrosome	(iv)	Graafian follicle

(1) a-iii, b-iv, c-i, d-ii
(2) a-iii, b-i, c-iv, d-ii
(3) a-i, b-iv, c-iii, d-ii
(4) a-iii, b-iv, c-ii, d-i

69) If for some reason, the vas deferens in the human reproductive system get blocked, the gamete will not be transported from :-

(1) Vagina to uterus
(2) Testes to epididymis
(3) Epididymis to Rete testis
(4) Epididymis to ejacultory duct

70) How many in the list given below constitute the female accessory ducts.
Oviducts, vagina, uterus, prostate, bulbourethral glands ?

(1) Four
(2) Three
(3) Two
(4) One

71) The head of a mature mammalian sperm consists of :-

(1) Acrosome and proximal centriole
(2) An elongated haploid nucleus
(3) Mitochondria only
(4) Distal centriole and an axial filament

72) Which of the following correctly describes the antrum ?

(1) Inner layer of tertiary follicle
(2) Outer layer of tertiary follicle
(3) Fluid filled cavity in tertiary follicle
(4) Fluid filled cavity in secondary follicle

73) Compare the given figure A with B and select the option which is correct for both.

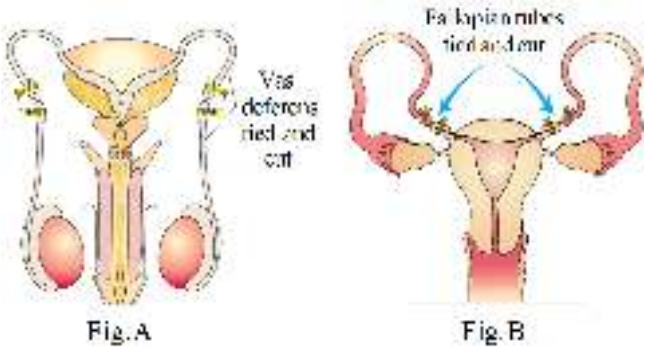


Fig. A Fig. B

(1) They prevent conception by blocking the entry of sperm through the cervix.
(2) Both are used for protecting users from contracting the sexually transmitted infections and AIDS
(3) These are one of the most widely accepted methods of contraception in in India
(4) Both are surgical methods of sterilisation, in the male is called "vasectomy" and in the female is called 'Tubectomy'

74) Which of the following is wrongly matched ?

(1) I.U.I. – Semen collected either from the husband or a healthy donor is artificially introduce in fallopian tube.
(2) I.C.S.I. – Sperm is directly inject into ovum
(3) G.I.F.T. – Transfer of ovum into the fallopian tube
(4) Z.I.F.T. – Zygote or early embryo (with up to 8 blastomere) is transferred into fallopian tube.

75) MTP is essential and helpful for

(1) Getting rid of unwanted pregnancies due to unprotected sex
(2) Getting rid of unwanted pregnancies due to failure of contraceptive used
(3) When continuation of pregnancy is harmful for mother or foetus or both
(4) All of these

76) Cu ions released from copper-releasing intra uterine devices (IUDs) :-

(1) Prevent ovulation

(2) Make uteres unsuitable for implantation

(3) Increase phagocytosis sperms

(4) Suppress sperm motility

77) Read the following statements A to E and select correct option-
(A) Big bang theory talks about singular huge explosion unimaginable in physical terms.
(B) The UV rays from the sun broke up water into hydrogen and oxygen.
(C) The universe expanded and hence temperature rise up.
(D) Miller observed formation of valine in Amino acids.
(E) Special creation theory has 4 connotations.

(1) A, B, C are correct

(2) C, D, E are correct

(3) A, B are correct

(4) A, C, D are incorrect

78) Given below are two statements :
Statement-I : Fossils are remains of hard parts of life forms found in rocks.
Statement-II : In 1938, a fish caught in South Africa happened to be a coelacanth which was thought to be extinct.

(1) Statement-I and Statement-II both are correct.

(2) Statement-I is incorrect but Statement-II is correct.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I and Statement-II both are incorrect.

79) A gene locus has two alleles. If the frequency of dominant allele is 0.4, then what will be the frequency of heterozygous individuals in the population which is in Hardy-Weinberg equilibrium?

(1) 0.24

(2) 0.6

(3) 0.36

(4) 0.48

80) The first non-cellular form of life could have originated about _____

(1) 3 million years ago

(2) 2 million years ago

(3) 3 billion years ago

(4) 2 billion years ago

81) When the same structure develops along different direction due to adaptation to different needs this is __X__ evolution and these structures are __Y__.
Choose the option which correctly fills the blanks

	X	**Y**

(1)	Divergent	Analogous
(2)	Convergent	Analogous
(3)	Convergent	Homologous
(4)	Divergent	Homologous

(1) 1

(2) 2

(3) 3

(4) 4

82) Fill in the blanks -
Our body when it encounter a pathogen for the first time produce response called __a__ response which is __b__ intensity. Subsequent encounter with same pathogen elict a __c__ intensified __d__ response.

(1) a-Anamnestic b-Low
c-Highly d-Secondary

(2) a-Secondary b-High
c-Low d-Primary

(3) a-Primary b-Low
c-Highly d-Anamnestic

(4) a-Primary b-High
c-Low d-Anamnestic

83) Which of the following statements about malaria are true?
(a) Malaria is caused by a bacterium called *Plasmodium vivax*.
(b) *Plasmodium falciparum* causes the most severe form of malaria.
(c) Malaria is transmitted by the bite of infected female *Anopheles* mosquitoes.
(d) The malaria parasite infects and ruptures red blood cells.
(e) Hemozoin is a toxic substance released due to rupture of red blood cells.
Choose the correct answer from the options given below :

(1) a, c, e only

(2) a, b, e only

(3) a, b, c, d only

(4) b, c, d, e only

84) Which statements is correct about given diagram ?  (a) Microsporum, Trichophyton and Epidemophyton are responsible for this.
(b) Allergic reactions
(c) Caused by wuchereria bancrofti
(d) Heat and moisture help these to grow

(1) a, c, d

(2) b, c, d

(3) a, d

(4) a, b, c, d

85) Given below are two statements:
Assertion A : Colostrum provides passive immunity to the infant.
Reason R : Colostrum secreted by the mother contains antibodies that protect the infant.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) **A** is correct but **R** is not correct.

(2) **A** is not correct but **R** is correct.

(3) Both **A** and R are correct and **R** is the correct explanation of **A**.

(4) Both **A** and R are correct but **R** is NOT the correct explanation of **A**.

86)

Match List I with List II :

List I		**List II**	
A.	Cocaine	I.	Effective painkiller
B.	Heroin	II.	*Cannabis sativa*
C.	Morphine	III.	*Erythroxylum*
D.	Marijuana	IV.	*Papaver somniferum*

Choose the correct answer from the options given below:
(1) A-IV, B-III, C-I, D-II

(2) A-I, B-III, C-II, D-IV

(3) A-II, B-I, C-III, D-IV

(4) A-III, B-IV, C-I, D-II

87) Which of the following correctly matches the organ with its role in the immune system?

(1) Spleen - filters blood and traps pathogen

(2) Thymus - Produce antibodies

(3) Thymus - Maturation of B-lymphocytes

(4) Lymph nodes - Produce red blood cells

88) which of the following cell directly attack virus infected cells without prior sensitization?

(1) B-lymphocyte

(2) Helper-T-cells

(3) T-killer cells

(4) Natural killer cells.

89) Which of following is given to cancer patients to helps in destroying the tumor ?

(1) Cannabinoids
(2) Anti retro viral therapy
(3) α - Interferon
(4) Lymphotoxin and antibiotics

90) HIV that causes AIDS, first starts destroying:

(1) N-K cell
(2) T-killer cell
(3) T-helper cell
(4) B-cell

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	3	2	2	1	3	3	4	4	2	3	4	1	3	1	4	4	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	1	4	3	3	2	3	3	4	4	4	3	2	2	1	4	2	3	4
Q.	41	42	43	44	45															
A.	3	1	4	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	3	1	2	4	2	1	3	2	4	2	1	4	3	1	3	4	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	1	1	1	1	2	4	2	4	1	3	1	1	1	4	2	3	2	2
Q.	86	87	88	89	90															
A.	2	3	4	1	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	3	1	4	3	3	1	3	4	4	2	4	4	1	1	3	1	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	3	2	2	2	1	1	2	1	4	4	2	3	1	2	2	1	1	4	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	1	3	1	3	2	4	2	3	3	2	2	3	1	1	4	2	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	4	4	4	4	2	1	4	2	2	3	4	1	4	4	3	1	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	3	4	3	3	4	1	4	3	3										

SOLUTIONS

PHYSICS

1) Magnetic field due to current carrying circular loop on its axis is given as

$$\frac{\mu_0 i r^2}{2(r^2 + x^2)^{3/2}}$$

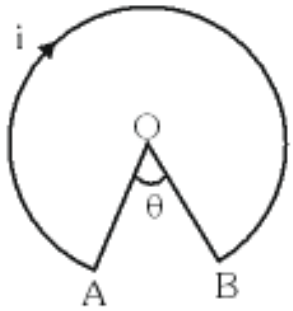
At centre, x = 0, $B_1 = \frac{\mu_0 i}{2r}$

At x = r, $B_2 = \frac{\mu_0 i}{2 \times 2\sqrt{2}r}$

$$\frac{B_1}{B_2} = 2\sqrt{2}$$

2) $B = \frac{\mu_0 i}{4\pi R'}(2\pi - \theta)$

where ; (2π – θ) R' = 2πR



$$R' = \frac{2\pi R}{2\pi - \theta}$$

$$B = \frac{\mu_0 i}{2R}\left(\frac{2\pi - \theta}{2\pi}\right)^2$$

3) $\vec{F} = I\left(\vec{\ell} \times \vec{B}\right) = 4\left(8\hat{i} \times 5\hat{k}\right) = -160\,\hat{j}\ N$

4) f = 50Hz

$X_L = 2\pi fL$

$= 2\pi(50)(200 \times 10^{-3})$

$= 20\pi\Omega$

$i_0 = \frac{V_0}{X_L} \Rightarrow \frac{V_{rms}\sqrt{2}}{X_L}$

$= \frac{(220)\sqrt{2}}{20\pi} = \frac{11\sqrt{2}}{\pi}$

$\boxed{i_o = \frac{\sqrt{242}}{\pi}} \Rightarrow a = 242$

5) Given $X_L = X_C = 5\Omega$, this is the condition of resonance. So $V_L = V_C$, so net voltage across L and C combination will be zero.

6) $P = \frac{I}{C} = \frac{uC}{C} = \frac{1}{2}\epsilon_0 E_0^2$

7)

UV rays are used for Lasik eye surgery. IR is used for physiotherapy. X-rays are used for cancer treatment. And Microwaves are used for aircraft navigation.

8)

Values given for coil :-
Number of turms of small coil = 10
Area of coil = 2.5 cm^2 = 2.5 × 10^{-4} m^2
$\Delta t = 25 \times 10^{-3}$ s
Values given for solenoid
 = 20 × 10^{-2} m, N = 240

$I = \frac{10}{\pi}Amp$, $n = \frac{240}{20 \times 10^{-2}} = 12 \times 10^2$

According to Faraday's law, induced emf $E = -\frac{d\phi}{dt}$
Induced emg in coil :-

$$I = \frac{\left|-\frac{d\phi}{dt}\right|}{R} = \frac{(NBA)/dt}{R} = \frac{\mu_0 nI(NA)}{\Delta tR}$$
$$= \frac{10 \times 4\pi \times 10^{-7} \times 12 \times 10^2 \times 10/\pi \times 2.5 \times 10^{-4}}{25 \times 10^{-3} \times 4.8}$$
$$= \frac{4 \times 12 \times 10^{-7} \times 10^2 \times 10 \times 10^{-4}}{10^{-3} \times 4.8} = 10^{-4}$$

I = 0.1 mA

9) After long time,
$I_A = \frac{2V}{7R} = 20mA$ .....(i)
Immediately after the key is closed,
$I = \frac{V}{3R} = \frac{70}{3}mA$

10)

due to high doping width of depletion layer decrease and strength of internal electric field increase

11) $Y = \overline{\overline{A.\overline{A.B}}\ .\ B + \overline{A.B}}$

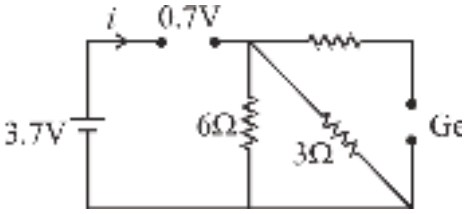
$= \overline{\overline{A.\overline{A.B}}.} + \overline{B + \overline{A.B}}$

$= A.(\bar{A} + \bar{B}) + \bar{B}.\overline{(A.B)}$

$= A.\bar{A} + A.\bar{B} + \bar{B}.A.B$

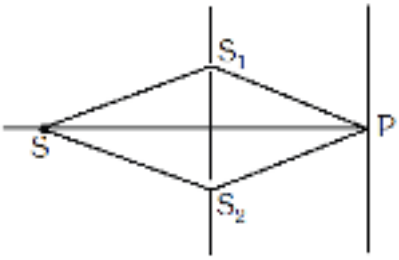
$= A.\bar{B}$

12)



$R_{eq} = 2\ \Omega$

$i = \frac{3.7 - 0.7}{2} = \frac{3}{2} = 1.5$ Amp

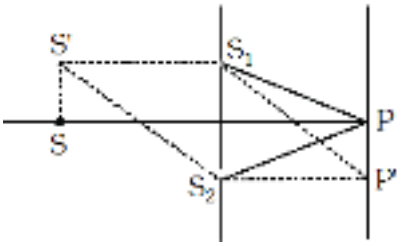


13)

Initially

$S\,S_1 + S_1\,P = S\,S_2 + S_2\,P$

Now the source is moved parallel to plane



Now $S'\,S_2 > S'\,S_1$

 $S_1\,P$ must increase so that path difference remains zero.

Hence the fringe width will remain same but fringe pattern will shift down.

14) Brewster's angle = 53°

$\tan 53° = \frac{n_d}{n_r} = \frac{4}{3}$

15)

Separation of negative electron from surface results in positive charge on it, which attracts that electron.

16) $\phi = h\upsilon_0$ No emission if $\upsilon < \upsilon_0$

$$\upsilon_0 = \frac{\phi}{h}$$

$$\upsilon < \frac{\phi}{h}$$

17) $E_3 - E_1 = 12.09$ eV (in H atom)

$eV_0 = h\upsilon - \phi_0$

$\phi_0 = (12.09 - 8.09) \times 1.6 \times 10^{-19}$

$$\upsilon_0 = \frac{4 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} \approx 10^{15} \text{ Hz}$$

18) COLM

$\vec{P}_i = \vec{P}_f$

$\vec{O} = \vec{P}_1 + \vec{P}_2$

$\vec{P}_1 = -\vec{P}_2$

$|\vec{P}_1| = |\vec{P}_2| = P$

$\lambda \propto \frac{1}{P}$

$$\frac{\lambda_1}{\lambda_2} = \frac{P_2}{P_1}$$

= 1

19) $Q = BE_f - BE_i$

$= 16 \times 7.97 - 17 \times 7.75$

= –4.23 MeV

So 4.23 MeV energy required

20) KE = –TE = 13.6 eV

PE = 2TE = 2 × (–13.6)

= – 27.2 eV

21)

When lens is in air,

From Lens maker's formula,

$$\frac{1}{f_1} = ({}^{a}\mu_g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{....(i)}$$

When lens is immersed in liquid,

$$\frac{1}{f_2} = \left({}^{\ell}\mu_g - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{...(ii)}$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{f_2}{f_1} = \frac{({}^{a}\mu_g - 1)}{({}^{\ell}\mu_g - 1)}$$

$$\Rightarrow \quad \frac{f_2}{f_1} = \frac{(1.5-1)}{\left(\frac{1.5}{1.25} - 1\right)} \quad \left(\because {}^{\ell}\mu_g = \frac{\mu_g}{\mu_\ell}\right)$$

($\mu_{\square}$ = 1.25)

$$\Rightarrow \quad \text{or } \frac{f_2}{f_1} = \frac{0.5 \times 1.25}{0.25} = 2.5 \qquad f_2 = 2.5\ f_1$$

Hence, focal length increases by a factor 2.5 when immersed in liquid.

22)

Since the angle of prism is very small
Therefore,

$$\frac{e}{r_2} = \mu$$

where $r_2 = A - r_1$

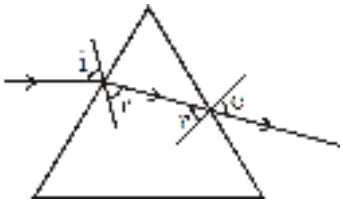
$r_2 = 3° - 0° = 3°$

$$e = \mu r_2 = \frac{3}{2} \times 3^o = 4.5°$$

23) $$\frac{d^1}{d} = \frac{\mu_2}{\mu_1}$$

$$\frac{d^1}{2} = \frac{5}{2} \Rightarrow d^1 = 5cm$$

24) Minimum deviation δ when
i = e
δ = A
$r_1 + r_2 = A \quad (r_1 = r_2 = r)$
2r = A
sini= μsinr



$$\text{sini} = \mu \sin\frac{A}{2}$$

δ = i + e – A
A = 2i – A
i = A

$$\sin A = \mu \sin\frac{A}{2}$$

$$2\cos\frac{A}{2} = \mu$$

$$\cos\frac{A}{2} = \frac{\mu}{2}$$

$$1 - \sin^2\frac{A}{2} = \frac{\mu^2}{4}$$

$$\sin^2\frac{A}{2} = 1 - \frac{\mu^2}{4}$$

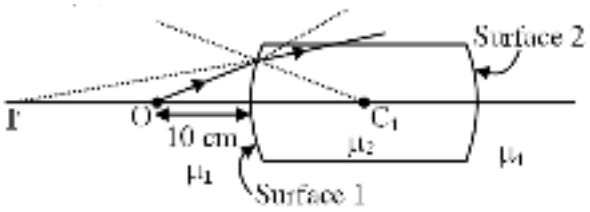
$$A = 2\sin^{-1}\sqrt{\frac{4-\mu^2}{4}}$$

25) $$v_{i/m} = -\left(\frac{v}{u}\right)^2 v_{O/m}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{60} - \frac{1}{40} \Rightarrow v = -120$$

$$v_{i/m} = -\left(\frac{120}{60}\right)^2 \cdot 10 = 40 cm/sec$$



26)

After first refraction, $u_1 = -10$ cm, $R_1 = 10$ cm

$$\frac{\mu_2}{v_1} - \frac{\mu_1}{u_1} = \frac{\mu_2 - \mu_1}{R_1}$$

$$\frac{3/2}{v_1} - \frac{1}{(-10)} = \frac{-3/2 - 1}{10}$$

$$\frac{3/2}{v_1} = \frac{1}{20} - \frac{1}{10} = \frac{-1}{20}$$

$v_1 = -30$ cm

first image I' will be form 30 cm left of surface 1. for second refraction → $u_2 = -50$ cm, $R_2 = -10$

$$\frac{\mu_1}{v_2} - \frac{\mu_2}{u_2} = \frac{\mu_1 - \mu_2}{R_2}$$

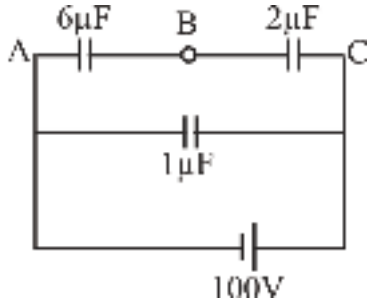
$$\frac{1}{v_2} - \frac{3/2}{(-50)} = \frac{1 - 3/2}{-10}$$, $v_2 = +50$ cm right of B.

27) $$V_{com} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

$$V_{common} = \frac{2C \times 2V - CV}{2C + C} \Rightarrow V_{common} = V$$

Final energy of configuration , $$U = \frac{1}{2}CV^2 + \frac{1}{2}(2C)V^2$$

$$= \frac{3}{2}CV^2$$



28)

$$V_{AB} = \frac{2}{2+6} \times 100V \; ; \; V_{BC} = \frac{6}{2+6} \times 100V$$

29) $$\int dQ = \int_0^3 I\,dt = \int_0^3 6\,t^2\,dt = 6\left[\frac{t^3}{3}\right]_0^3 = 54 \text{ C}$$

$$R = \frac{\rho\ell}{A} = \frac{\rho\ell}{\left(\frac{M}{\ell d}\right)} = \frac{\rho\ell^2 d}{M}$$

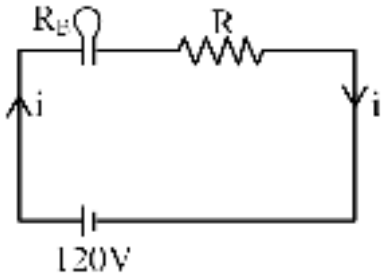
30)

$$R \propto \frac{I^2}{m} \Rightarrow R_1 : R_2 : R_3 = \frac{I_1^2}{m_1} : \frac{I_2^2}{m_2} : \frac{I_3^2}{m_3}$$

$$\Rightarrow R_1 : R_2 : R_3 = \frac{9}{1} : \frac{4}{2} : \frac{1}{3} = 27 : 6 : 1$$

31)

Circuit is



$$R_{lamp} = \frac{V^2}{W} = \frac{(30)^2}{90} = 10\Omega$$

$$i = \sqrt{\frac{W}{R}} = \sqrt{\frac{90}{10}} = 3A$$

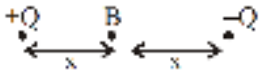
$$i = \frac{V}{R + R_B} \Rightarrow 3 = \frac{120}{R + 10} \Rightarrow R + 10 = 40$$

$$\Rightarrow R = 30\ \Omega$$

32) 

$E_A = 0$

$V_A \neq 0$



$E_B \neq 0$

$V_B = 0$

33)

Electron will follow projectile motion for projectile $H_{max} = \frac{u^2 \sin^2\theta}{2g}$

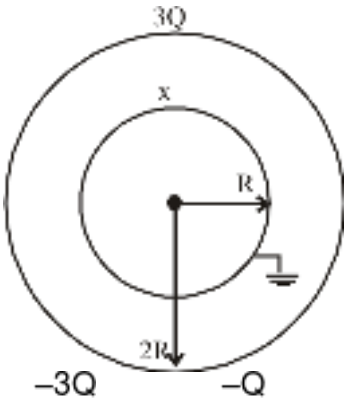
$$d = \frac{u^2\left(\frac{1}{\sqrt{2}}\right)^2}{2 \times \text{acceleration}} \qquad \left[\because\ a = \frac{qE}{m}\right]$$

$$d = \frac{u^2 m}{2 \times 2 \times qE} \Rightarrow d = \frac{k}{2eE} \text{ or } E = \frac{k}{2ed}$$

34) When inner shell is grounded the potential of inner shell becomes zero because potential of the Earth is taken to be zero.

$$\frac{Kx}{R} + \frac{K3Q}{2R} = 0$$

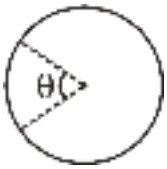
$x = \frac{-3Q}{2}$, the charge that has increased



$= \frac{-3Q}{2} - (-Q) = \frac{-Q}{2}$.

Hence the amount of charge that flows into the Earth = $\frac{Q}{2}$

35) Electric field due to arc

at its centre = $\frac{2k\lambda}{R}\sin\left(\frac{\theta}{2}\right)$ 

Applying principle of superposition

$$\vec{E}_{net} = \vec{E}_{arc} + \vec{E}_{remaining} = 0 \Rightarrow E_{remaining} = -\vec{E}_{arc}$$

$$\left|\vec{E}_{remaining}\right| = \frac{1}{4\pi\varepsilon_0 R}\cdot\frac{Q}{2\pi R}\cdot 2\sin\left(\frac{\theta}{2}\right)$$

$$= \frac{Q}{4\pi^2\varepsilon_0 R^2}\sin\left(\frac{\theta}{2}\right)$$

36) • Initial gap d, thickness t = 0.6d, effective gap d_{eff} = d – 0.6d = 0.4d.

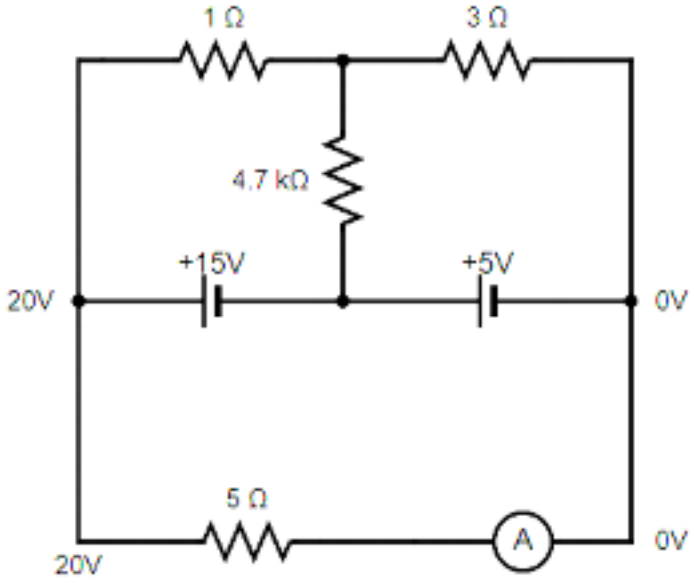
• Cinitial = $\frac{C}{0.4} = \frac{20}{0.4} = 50\ nF$, C_{final} = 20 nF.

• $W_{mech} = \frac{1}{2}(C_i - C_f)V^2$.

• $W_{mech} = 0.5 \times (50 - 20) \times 10^{-9} \times 100^2$

$= 1.5 \times 10^{-4}$J.

• W_{mech} = 150 µJ.



37)

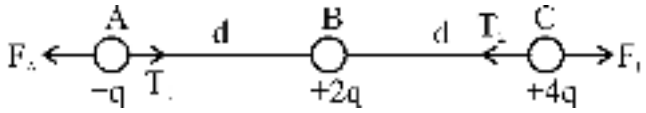
From the concept Diagram voltage drop across 5Ω resister is

$V = 20 = (i_A).(R)$

$20/5 = (i_A)$

$i_A = 4$ A

38)



$$T_1 = k\left[\frac{2q^2}{d^2} + \frac{4q^2}{4d^2}\right] = \frac{3kq^2}{d^2}$$

$$T_2 = k\left[\frac{8q^2}{d^2} + \frac{4q^2}{4d^2}\right] = \frac{9kq^2}{d^2}$$

$$\frac{T_1}{T_2} = \frac{1}{3}$$

39)

$p = vT$

- The time period of circular motion is:

$$T = \frac{2\pi m}{qB}$$

- Since $v = \frac{v}{\sqrt{2}}$ we substitute:

$p = \frac{v}{\sqrt{2}} \cdot \frac{2\pi m}{qB}$; $v = \frac{pqB}{2\pi m}\sqrt{2}$ and $r = \frac{mv}{\sqrt{2}qB}$

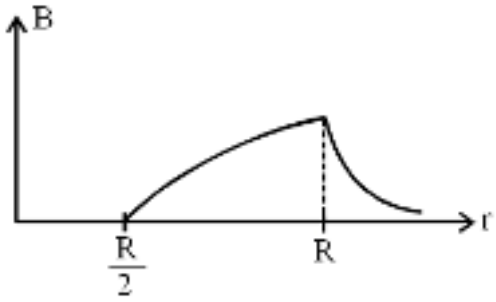
Substituting v:

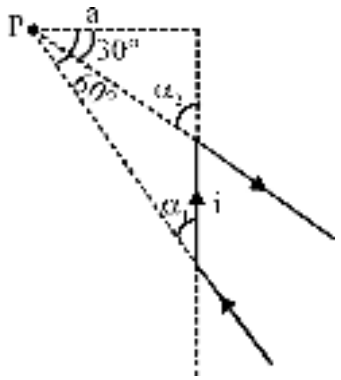
$r = \frac{m}{\sqrt{2}qB} \times \frac{pqB}{2\pi m}\sqrt{2}$, $r = \frac{p}{2\pi}$

40) $r < \frac{R}{2}$, $B = 0$

$r > R$, $B = \frac{\mu_0 I}{2\pi R}$

$\frac{R}{2} < r < R$, $B = \left[\frac{r^2 - \left(\frac{R}{2}\right)^2}{R^2 - \left(\frac{R}{2}\right)^2}\right]$





41)

$B_p = \frac{\mu_0 i}{4\pi a}(\sin 60° - \sin 30°)$

$= \frac{\mu_0 i}{8\pi a}\left(\sqrt{3} - 1\right) \odot$

42)

$Bi\ell = mg$

$B\left(\frac{Bv\ell}{R}\right)\ell = mg$

$v = \frac{mgR}{B^2\ell^2}$

43)

From the graph :
At V = 0.6 V, I = 50 mA
At V = 0.8 V, I = 60 mA
So, ΔV = 0.8 – 06 = 0.2V
ΔI = 60 – 50 = 10 mA = 0.01A

$$r_d = \frac{0.2}{0.01} = 20\Omega$$

44)

 50 VSD = 48 MSD

 1 VSD = $\frac{48}{50}$ MSD = 0.96 MSD

= 0.96 × (0.05 cm) = 0.048 cm

So by the formula,

LC = (1 MSD − 1 VSD)

= (0.05 cm − 0.048 cm) = 0.002 cm

= 2×10^{-3} cm

Given value in the question,

LC = $x \times 10^{-3}$ cm

On comparing these two values of LC, we get -

x = 2

45) $R = \frac{V}{I} = \tan 53^o = \frac{4}{3}\Omega$; r = 10^{-3} m

$$\frac{4}{3} = \frac{\rho\ell}{A} \Rightarrow \rho = \frac{4}{3}\frac{A}{\ell} = \frac{4}{3}\frac{\left(3.14 \times 10^{-6}\right)}{(0.5)}$$

$\rho \approx 8 \times 10^{-6}$ Ω-m

CHEMISTRY

46)

Slope = $\frac{-E_a}{2.303R} = -\frac{1}{2.303} \Rightarrow E_a = R = 2$ cal

47) $2SO_2 + O_2 \rightarrow 2SO_3$

$$+\frac{1}{2}\frac{d[SO_3]}{dt} = \frac{-d[O_2]}{dt} = \text{R·O·R}$$

$$\frac{1}{2}\left(\frac{0.8}{80}\text{mol. lit}^{-1}\text{ sec}^{-1}\right) = \frac{d[O_2]}{dt}$$

$$\frac{1}{200}\text{mol. lit}^{-1}\text{ sec}^{-1} = \frac{d[O_2]}{dt}$$

Now $\frac{d[O_2]}{dt} = \frac{1}{200} \times 32$ g.lit^{-1} sec^{-1}

⇒ 0.16 g.lit^{-1} sec^{-1}

48) $\frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{P_2}{P_1}\right)^{n-1}$

$$\frac{446}{669} = \left(\frac{182}{273}\right)^{n-1}$$
$$\frac{2}{3} = \left(\frac{2}{3}\right)^{n-1}$$
n – 1 = 1
n = 2

49) From given data $t_{1/2}$ = 30 minute
$$K = \frac{\ln 2}{30}$$
$r_{t=60} = k[A]_{t=60}$
$$= \frac{\ln 2}{30} \times \frac{0.05}{2} = \frac{\ln 2}{1200}$$

50) At $t_{1/8} \Rightarrow x = \frac{[A]_0}{8}$ $x = Kt$
$$\frac{[A]_0}{8} = Kt_{1/8} \quad ...(1)$$
At $\frac{t_{15}}{16} \Rightarrow x = [A]^0\left(\frac{15}{16}\right)\frac{15}{16}[A]_0 = K \times \frac{t_{15}}{16}$ ...(2)
$$\frac{\text{eq. (1)}}{\text{eq. (2)}} = \frac{\frac{t_{1/8}}{\frac{t_{15}}{16}}}{\frac{\frac{A_0}{8}}{\frac{15}{16}A_0}} = \frac{16}{15} \times \frac{1}{8} = 2:15$$

51) $$\frac{w_1}{w_2} = \frac{Z_1 i_1 t_1}{Z_2 i_2 t_2} = \frac{E_1 i_1 t_1}{E_2 i_2 t_2}$$
$w_1 = w_2$, $t_1 = t_2$ } Given $1 = \frac{E_1}{E_2}\frac{i_1}{i_2}$ $\frac{i_1}{i_2} = \frac{E_2}{E_1}$

52) ⇒ n-factor = 2
gm eq. (C_2H_6) = Faraday
$$\text{Moles} \times \text{V.F.} = \frac{it}{96500}$$
$$\left(\frac{V_L}{22.4}\right) \times 2 = \frac{0.5 \times \frac{80}{100} \times 965 \times 60}{96500}$$
VL = 2.688 L

53)
$$E_{cell} = E^0_{cell} - \frac{0.0591}{2} \times \log \frac{[H^+]^2}{[Ni^{2+}]p_{H_2}}$$
$$\Rightarrow E_{cell} - E°_{cell} = \frac{0.0591}{1} \log [H^+] = 0.0591 \times pH$$
⇒ pH = 1

$H_2 \rightarrow 2H^+ + 2e^-$
$Ni^{2+} + 2e^- \rightarrow Ni(s)$
$H_2 + Ni^{2+} \rightarrow 2H^+ + Ni(s)$.

54)

$(SRP)_{D^+/D} < (SRP)_{A^+/A}$
 D is more reactive than A and can displace A^+.

55) $(\Lambda_m^0)_{NaNO_3} = (\Lambda_m^0)_{NaCl} + (\Lambda_m^0)_{AgNO_3} - (\Lambda_m^0)_{AgCl}$
= 110.3 + 116.5 – 121.6
= 105.2 S cm^2mol^{-1}

56) $\left(200 = \frac{P_i}{4} + \frac{P_P \times 3}{4}\right)$ ....(1)
$\left(300 = \frac{3P_i}{4} + \frac{P_P}{4}\right) \times 3$ ....(2)
$700 = \frac{9}{4}P_i - \frac{P_i}{4} = 2P_i$
$P_i = \frac{700}{2} = 350$ mm of Hg

57) $\pi_1 = \pi_2$
$i_1c_1 = i_2c_2$
$(1 + 2\alpha_1) \times \frac{17.4 \times 1000}{174 \times 100}$
$= (1 + \alpha_2) \times \frac{5.85 \times 1000}{58.5 \times 100}$ ($\alpha_2 = 1$)
$\alpha_1 = 0.5$ $\therefore \alpha$ in % = 50%

58) $\Delta T_b = i.\ k_b\ .\ m$
$1.518 = i \times 2.53 \times 1$
$i = 0.6$
 $i = 1 + \left(\frac{1}{n} - 1\right)\alpha$
$0.6 = 1 + \left(\frac{1}{2} - 1\right)\alpha$ (n = 2 ; for dimer)
$\alpha = 0.8 = 80\%$

59)

$C = \frac{(2.22/111)}{(100/1000)} = 0.2$
$\pi = iCRT$
$= 3 \times 0.2 \times 0.08 \times 300$
= 14.4 atm

60) $i = 1 + \left(\frac{1}{n} - 1\right)\alpha$

for X $i = 1 + \left(\frac{1}{2} - 1\right) 0.8 = 0.6$

for Y $i = 1 + \left(\frac{1}{3} - 1\right) 0.9 = 0.4$

$i \propto B.P. \propto \frac{1}{F.P.}$

61) NO_2 is a reddish-brown gas.

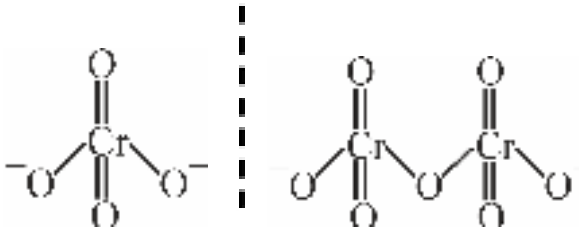
62) CaC_2, Al_4C_3 and $Be_2C \Rightarrow$ Ionic
SiC ⇒ Covalent network

63) $H\overset{+3}{N}O_2$, $H_2\overset{-1}{O}_2$, $\overset{+4}{S}O_2$ } They all behave as oxidising and reducing agent

64) $XeF_6 \xrightarrow{\text{excess } H_2O} 6HF + XeO_3$

$XeF_6 \xrightarrow{2H_2O} 4HF + XeO_2F_2$

65) Starting material for the preparation of potassium dichromate is $FeCr_2O_4$.

66) 

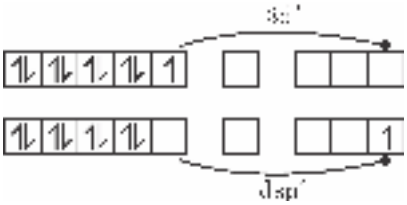
$Na_2Cr_2O_7$ is not used as a primary standared in volumatry due to its moisture absorbing nature.

67) Transition elements show variable oxidation states differing by one unit.

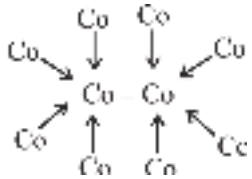
68) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} Cr_2O_3 + N_2 + H_2O$
green

69) Both $[Ni(CO)_4]$ and $[Ni(NH_3)_4]^{2+}$ have zero unpaired e^-.

70) $Cu^{2+} \rightarrow 4s^{\circ}3d^{9}$



71)



Co = Z → 27

Number of co-ordinate bond → 4

Number of delta bond → 1

So value of 'n' in $(Co)_n$ – Co – Co – $(Co)_n$ = 4

72) [M(A)(B)(C)(D)] square planar complex shows three geometrical isomers.

73) Cl^- ion gives positive chromyl chloride test

$Fe^{+3} + K_4[Fe(CN)_6]$® $Fe_4[Fe(CN)_6]_3$

Prussian blue

74)

Explain Question:

We are asked: Which salt in acidic solution produces a deep blue colour with starch-iodide solution?

Concept:

Starch-iodide test detects free iodine(I_2).

Certain ions (like nitrite) act as oxidising agents in acidic medium, liberating iodine from iodide.

Liberated iodine reacts with starch → deep blue colour.

Solution:

Chloride → does not liberate iodine

Nitrite → in acidic medium : $2NO_2^- + 2I^- + 4H^+ \rightarrow 2NO + I_2 + 2H_2O$ → iodine liberated → deep blue with starch

Acetate → no reaction

Bromide → does not liberate iodine under these conditions.

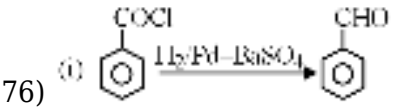
Final answer: Option (2)

75) $Fe^{+2} + [Fe(CN)_6]^{-4} \rightarrow$ White ppt.

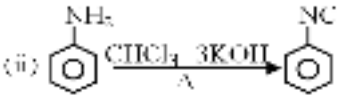
$Fe^{+3} + [Fe(CN)_6]^{-4} \rightarrow$ Prcessian blue ppt.

$Zn^{+2} + [Fe(CN)_6]^{-4} \rightarrow$ Bluish white ppt.

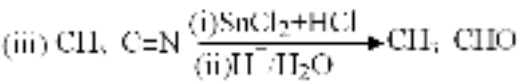
$Cu^{+2} + [Fe(CN)_6] \rightarrow$ Chocolate brown ppt.

76) 

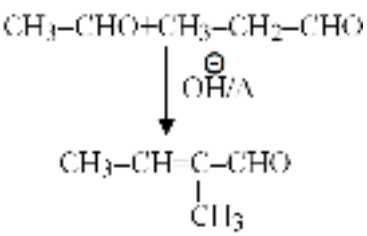
(i)



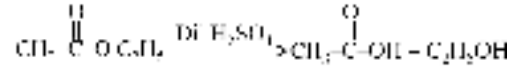
(ii)



(iii)

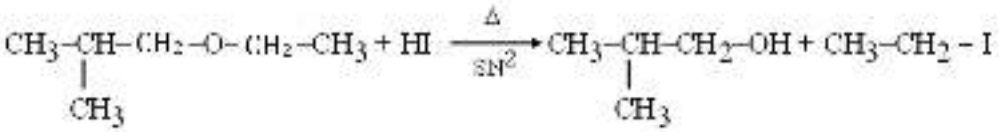


(iv)

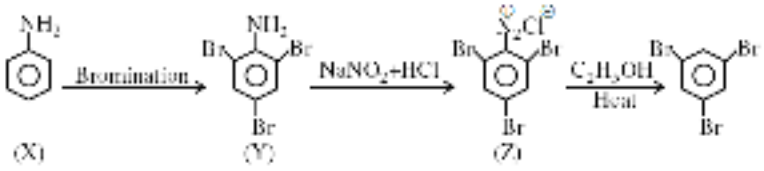
77) 

	CH_3COOH	C_2H_5OH
$NaHCO_3$	✓	✗
H.V.Z Rxn	✓	✓
Cl_2 + NaOH	✓	✓
Cu/300°	✗	✓

78) **Solution :** Dissociation of ether by acid (HI) is favored by S_N^2 mechanism.

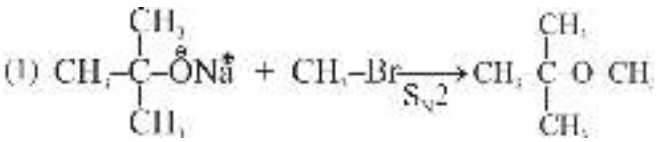


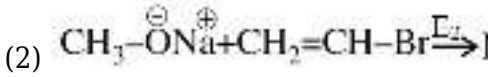
Final answer : Option (1)

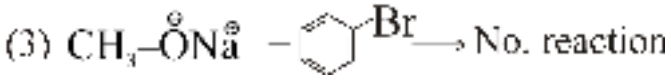
79) 

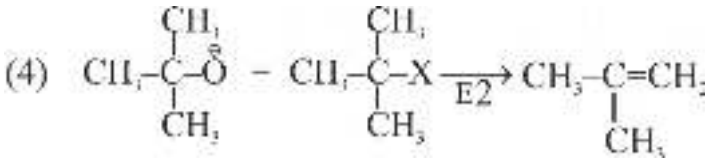
80)

Solution:

(1) 

(2) 

(3) 

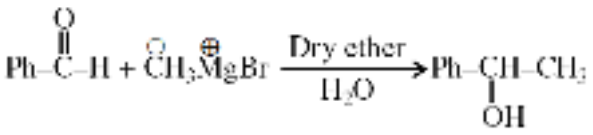
(4) 

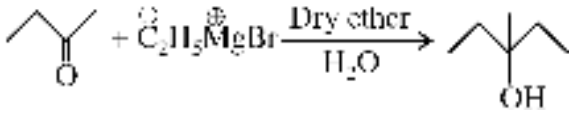
Final Answer: Option (1)

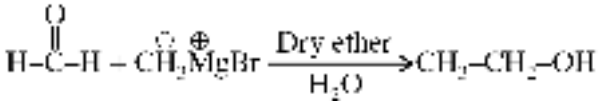
81)

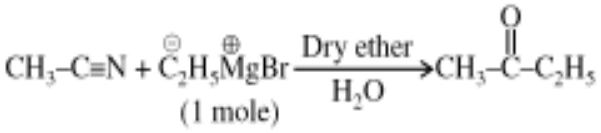
Rate of N.A.R ∝ electrophilicity of carbon of carbonyl group ∝ –M, –H, –I

82)

(I) 

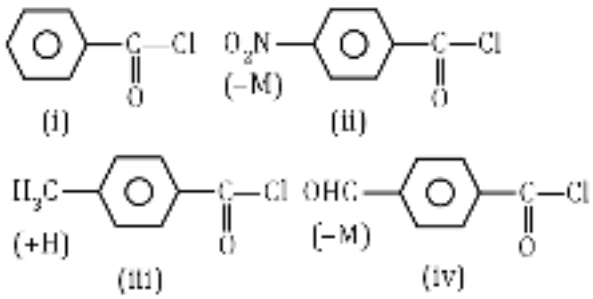
(II) 

(III) 

(IV) 

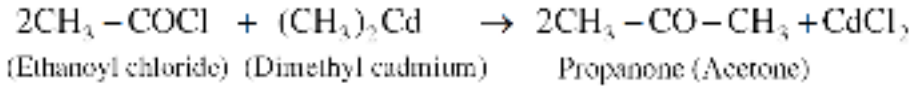
83)

Solution :- Reactivity towards SNAE ∝ Electrophilicity of carbonyl carbon.
∝ +ve charge on casrbonyl carbon

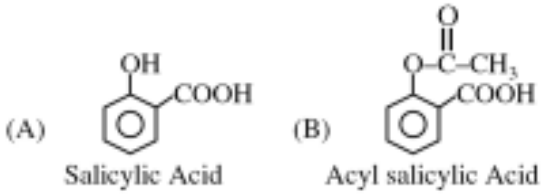


During hydrolysis, H_2O attacks as a nucleophile on carbonyl group. If group attached to carbonyl group, are electron withdrawing, it increases the electrophilicity of carbonyl carbon and hence increases its tendency to react with $Nu^{\ominus}$.
Therefore, the order will be(ii) > (iv) > (i) > (iii).
Final answer :- (3)

84) **Solution:** Grignard reagents (CH_3MgCl) are highly reactive and convert acid chlorides to 3° alcohols (A) by double addition.
Organocadmium reagents ($(CH_3)_2Cd$) are less reactive and stop at the ketone stage (B).
Organometalliccompounds – Grignardreagent

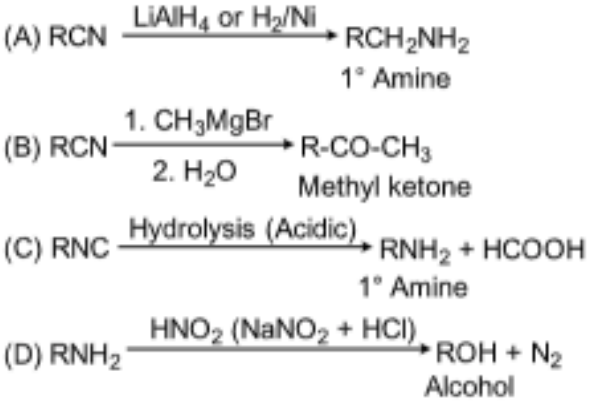


Final Answer: Option (2)



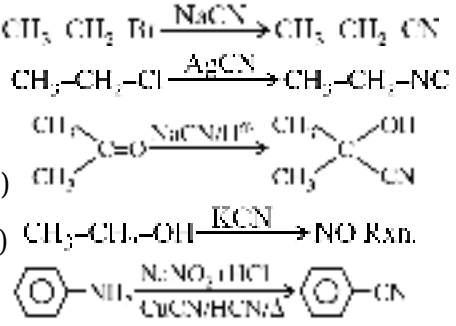
85)

86) **Solution:**

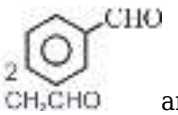


Final answer: option (2)

87)



88) **Solution :**

	Pairs of compounds	Lab test	Remark
1	Methyl Alcohol and Ethyl Alcohol	I_2 + Base (Iodoform Test)	Methyl alcohol does not gives positive iodoform test (yellow precipitate of CHI_3) Ethyl alcohol gives positive iodoform test (yellow precipitate of CHI_3) ⇒ Differentiates given compounds.
2	 and	Fehling Solution	Fehling's solution oxidizes CH_3CHO (aliphatic aldehyde) → brick red ppt Benzaldehyde does not give Fehling's test ⇒ Differentiates given compounds.

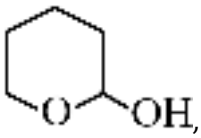
3	Phenol and Carboxylic Acid	$NaHCO_3$	Carboxylic acid reacts with $NaHCO_3 \rightarrow$ effervescence (CO_2 gas) Phenol does not react with $NaHCO_3$ $\Rightarrow$ Differentiates given compounds.
4	Propan-1-amine and Propan-2-amine	Hinsberg reagent	Both are 1°amine forms soluble sulfonamide. $\Rightarrow$ Given compounds do not differentiate.

Final answer : (4)

89)

I, III — N.S.R II, IV — F.S.R V — E.S.R

90)

(Hemi Acetal)  , Terminal Alkyne & α-Hydroxy ketone & Aldehyde give silver mirror with Tollen's Reagent.

BIOLOGY

91)

NCERT XI, Pg. # 59

92)

Explain Question : The question asks to identify the plant organ that shows polyarch condition, has a large and well:developed pith, and does not undergo secondary growth.

Concept : Anatomy of Flowering Plants : Anatomy of Roots – Monocot and Dicot Roots

Solution : Monocot roots show polyarch condition (many xylem and phloem bundles), have a large and well:developed pith, and do not undergo secondary growth due to absence of vascular cambium.
Dicot roots usually have fewer xylem bundles and undergo secondary growth, while stems do not show polyarch condition.
Final answer : 2

93)

NCERT XI, Pg. # 59

94)

NCERT XI, Pg. # 66

95)

The correct answer is:
Both Statement-I and II are correct
• **Statement-I:** "The shape, margin, apex, surface, and extent of incision of lamina vary in different leaves. 11
• Correct. The st!ucture of leaves, including shape, margin, apex, surface texture, and the depth of incision (indentation in the leaf edge) can vary significantly across different plant species.
• **Statement-II:** "Veins provide rigidity to the leaf blade and act as channels of transport for water, minerals, and food materials."
• Correct. The veins in leaves help maintain the structural rigidity of the leaf and are essential for transporting water, minerals, and food materials (like sugars) throughout the leaf.

96)

Explanation :

Step 1: Identify Ovary Position for Each Plant

Plant	**Ovary Position**	**Superior/Inferior**
China Rose (Hibiscus)	Superior	□
Mustard (Brassica)	Superior	□
Brinjal (Solanum)	Superior	□
Potato (Solanum)	Superior	□
Guava (Psidium)	Inferior	□
Cucumber (Cucumis)	Inferior	□
Onion (Allium)	Superior	□
Tulip (Tulipa)	Superior	□

Step 2: Count the Plants with a Superior Ovary

A. Superior ovary is found in China Rose, Mustard, Brinjal, Potato, Onion and Tulip → Total = 6

Final Answer: (3)

97)

NCERT XI, Pg. # 74

98)

NCERT XI, Pg. # 72

99)

NCERT XII, Pg. # 12

100)

NCERT XII, Pg. # 7

101)

NCERT XII, Pg. # 10

102)

NCERT XII, Pg. # 7

103)

NCERT XII, Pg. # 13

104)

NCERT XII, Pg. # 12

105)

NCERT XII, Pg. # 73

106)

NCERT XII, Pg. # 73

107)

NCERT XII, Pg. # 74

108)

NCERT XII, Pg. # 70,71

109)

Question Explanation: Identifying autosomal recessive disorders.

Concept: Autosomal recessive inheritance

Solution:

An autosomal recessive disorder requires two copies of the mutated gene (one from each parent) to express the trait. From the list, Cystic fibrosis, Phenylketonuria, Thalassemia, and Sickle cell anaemia are all autosomal recessive disorders. Haemophilia is X-linked recessive, and Myotonic dystrophy is an autosomal dominant disorder. Therefore, there are four such disorders.

So, the correct answer is 4 or option (1)

Final Answer: Option (1)

110)

NCERT - XII, Pg. # 89

111)

NCERT XII, Pg. # 61

112)

NCERT XII, Pg. # 83

113)

NCERT XII, Pg. # 83

114)

NCERT XII, Pg. # 89

115)

NCERT XII, Pg. # 83

116)

Explain Question : The question asks about the contributions of Severo Ochoa and Hargobind Khorana in RNA synthesis.

Concept : Genetic Code

Explanation :

Statement-I: Severo Ochoa developed polynucleotide phosphorylase for enzymatic synthesis of RNA → Correct

Statement-II: Hargobind Khorana developed the chemical method for RNA synthesis → Correct

Final answer : Option (1)

117) **Explain Question :**The question asks how many of the given statements about RNA are correct.

Concept : RNA, Types Of RNA

Explanation :
Statement I: True — single RNA polymerase transcribes all RNA types in bacteria.

Statement II: True — the three major RNA types are correct.

Final answer : Option (1)

118) NCERT Pg.#80,96

119) NCERT Pg.# 109.

120) NCERT XII, Pg.# 108 (E), 118-119 (H)

121) NCERT Pg. # 207

122)

NCERT XII Pg # 207

123)

NCERT XII Pg # 210

124) NCERT Page #205

125) NCERT Pg. No.#206,207,211.

126)

NCERT XI Pg # 198-202

127)

NCERT XII Pg # 198-202

128) NCERT XII Pg # 199
Explanation: Competitive release happens when a species restricted by competition expands its range once the competitor is removed.

129)

NCERT XII Pg # 199

130)

NCERT XII Pg # 212

131) NCERT Pg. No. # 220

132) NCERT Pg. No. # 225

133) NCERT Pg.#265

134) NCERT Pg.#265

135) NCERT Pg.#221,222,223

136)

NCERT XII, Pg. # 171

137) NCERT XII, Pg. # 168

138) NCERT XIIth Pg # 199

139) NCERT Pg # 199

140) NCERT 12 Pg. 172

141) NCERT 12th, Page No. # 165

142) NCERT Pg.# 169

143) NCERT-XII, Pg. # 179 (E), 196 (H)

144) NCERT, Pg. # 232

145) NCERT, Pg. # 184

146) NCERT XII, Pg. # 208, 209, 212, 213 (E)
228, 229, 232, 233, 234 (H)

147)

NCERT-XII, Pg. # 214

148) NCERT-XII, Pg. # 182

149) NCERT Pg # 154

150) NCERT Pg. # 151, 152 (E), 166, 167 (H)

151) NCERT, Pg. # 155, 156

152) NCERT Pg. # 153

153)

NCERT, Pg. # 38

154) NCERT-XII, Pg. # 38, Para-2.7

155) NCERT PAGE (ENGLISH): 51
NCERT PAGE (HINDI): 55

156) NCERT, Pg. # 27

157) NCERT, Pg # 37

158) NCERT Pg. # 29

159)

NCERT, Pg. # 27

160)

NCERT, Pg. # 29

161) NCERT-XII, Pg. # 32, Para-2.3

162)

NCERT, Pg. # 32, 33

163) NCERT Pg. # 45

164) NCERT Pg.# 48

165)

NCERT, Pg. # 46

166)

NCERT, Pg. # 44

167) NCERT Pg # 110-113

168)

NCERT, Pg. # 113,122

169)

NCERT, Pg. # 121

170)

NCERT, Pg. # 111

171)

NCERT, Pg. # 114,115

172)

NCERT, Pg. # 135

173)

NCERT, Pg. # 131,132

174)

NCERT, Pg. # 133

175)

NCERT, Pg. # 136

176) NCERT-XII, Pg. # 142, 143

177)

NCERT, Pg. # 138

178) NCERT, Pg. # 134,135

179)

NCERT, Pg. # 142

180)

NCERT, Pg. # 139,140